Republic of Tunisia Ministry of Higher Education and Scientific Research

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University of Sousse

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Higher School of Science and Technology Hammam Sousse

Licence in Electronics Electrotechnical and Automation

Speciality: Embedded Systems

END-OF-STUDIES INTERNSHIP REPORT

Automation and supervision of the Blow room line "BDT-AXIFLOW-MPM-CVT4-DX"

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Academic Year: 2023-2024

Gratitude

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General Introduction

Nowadays, industrial automation is a factor of competitiveness and technological monitoring. The goal of automation is to reduce manual operator tasks and replace them with automatic tasks to improve product quality, reduce waste, improve operator and machine safety, and reduce cost. In addition, new human-machine interface technologies have been developed to monitor the status of industrial facilities in real time and to be able to inform themselves more quickly of any faults and incidents. Thanks to these solutions, intervention times are greatly reduced and anomalies can be taken into account.

It is in this context that our end-of-studies internship at the industrial textile company SITEX is taking place, the objective of which is the automation and supervision of a Blow room line "BDT-AXIFLO-MPM-CVT4-DX" .

The results of our work are presented in this report, which is organized as follows:

- The first chapter is devoted to the presentation of the host company.
- The second chapter contains the specification and detailed presentation of the Blow room line.
- The third chapter contains the elaboration of the various Grafcets describing the operation of the Blow room machines.
- The fourth chapter contains the selection of automation components (PLC and HMI) and their TIA programming and the implementation of the machine control and supervision program.
- The report will conclude with a general conclusion

Chapter 1: Company Overview

1. Introduction

In order to put the project in its content, it is essential to present the host company SITEX ; its activities and the production cycle. And then we're going to present his specifications.

2. Company Overview

2.1 Activities of the SITEX group and history



Figure 1: Logo Sitex

SITEX is a Tunisian textile industrial company that was created on 7 January 1977 as part of the restructuring of SOGITEX. It is specialized in the production and marketing of a wide range of denim fabrics dedicated mainly to export, Also, it produces spools of cotton thread according to different characteristics intended for the manufacture of jeans and various sports wear.
It has taken the top spots in Africa and the fourth place internationally in terms of product quality and quantity, which has allowed it to invade the jeans manufacturing market.

2.2 SITEX Branches

This company is made up of two geographically isolated units.

2.2.1 Sousse Unit

The Sousse SITEX unit is located in the city of Tafalla. It is spread over an area of 80464 m2. It is intended for the production of raw cotton yarn spools from imported cotton in accordance with the needs of the Ksar Helal mill (see figure below). This unit produces about 15000 tons of cables each year.

Figure 2: SITEX Sousse Unit

2.2.2 Ksar Hellal Unit

The Ksar Hellal unit takes care of the production of weaving, i.e. the dyeing, weaving and finishing of the fabric according to the desired characteristics. This factory is spread over an area equal to 1714944m2. It is equipped with a co-generation plant that simultaneously delivers the thermal and mechanical energy necessary for the proper functioning of the plant.

Figure 3: SITEX Ksar Unit

2.3 Share capital

The industrial textile company SITEX has a share capital of approximately 23,063,300 Dinars fragmented into 2,306,330 shares of 10 Dinars each as of November 31, 2020 as shown in the table below

Table 1: Company shareholders

2.4 Key figures

The key figures of the actions carried out by SITEX are given on the histogram in the figure below

Figure 4: Key figures

According to the histogram above, there is an increase in net sales in 2019 compared to 2017. On the other hand, operating profit decreased in 2019 after a change in 2018. In addition, net income and earnings per share were slightly higher

2.5 Production cycle of the Sousse unit

Since our end-of-studies internship was carried out in the SITEX company, Sousse unit, in this part, we will present the production cycle of cotton thread spools. The spinning process is the set of operations carried out in order to obtain fibre nets

Figure 6: Production Cycle

Can be used by processing cottons that are presented in bale form. This results from the transition of raw materials, which is cotton, from a fibrous state to a yarn of constant quality. The production cycle is carried out according to the steps shown in the following figure:

As shown in the figure above, the different stages of the production cycle of cotton yarn spools are:

- **Raw materials store** : dedicated to the storage of imported cotton that arrives at SITEX in the shape of bales. This material comes from several countries such as Senegal, Greece, Burkina Faso...
- **Threshing** : this action ensures the cleaning and homogenization of cotton of all types of waste using the machines: Blendomat BDT, Pre-cleaning Axiflo,Multi-mixing MPM, Cleanomat CVT4 and Dedusting DX to obtain a high quality cotton.
- **Carding** : which cleans the fibre and transforms it into a ribbon through the carding.

Figure 7: Carding Step

- **Stretching** : which allows the ribbons of the cards to be stretched to ensure good quality and a well-adjusted thickness.

Figure 8: Streching Step

- **Combing** : this step increases the characteristics of the yarn in terms of fineness, regularity and appearance.

Figure 9: Combing Step

- **Open-end** : This is a spinning operation that transforms the spindles into well-wound spools.

Figure 10: Open-end Step

- **Spindle bench** : this is the second pass of the stretching, its role is to transform the ribbons into strands that will then be wound on tubes and then sent to the CAF which transforms the wick into a wire with a precise fineness in a fully automatic way.



Figure 11: Spindle bench Step

- **Winding** : This task winds the wires on a cylindrical support.

Figure 12: Winding Stage

- **Assembly and Twisting** : This operation requires an assembly of two single threads to ensure a low twist.

Figure 13: Assembly and Twisting Step

- **Fixation by spray** : This step develops the twisting of the wire by water vapour in order to avoid the inverter appearance of the wires using a heat treatment machine.

Figure 14: Spray Fixation Step

- **Warping** : Threads are wound on a large cylinder for the Hellal ksar factory

Figure 15: Warping step

2.6 Flow chart

SITEX has several departments, which are headed by the General Manager as shown in the organizational chart shown in the following figure:

Figure 16: Organizational chart of senior management

3. Specifications

The specifications proposed to us by the company SITEX Sousse unit are based on the automation and supervision of the Blow room line machines with a Siemens S7 type PLC.

The objective of this task is to improve the performance of the machines and guarantee great flexibility in the management of the machines' operation.

The specifications of the task to be carried out in our end-of-studies internship are organized as follows:

- ❖ Study of the operation of the existing machines: determination of the different tasks as well as all the sensors, actuators involved in the operation of the machine
- ❖ Creation of the different Grafcets describing the operation of the machines: Management Grafcet and Safety Grafcet.
- ❖ Implementation of the program on the Siemens S7 controller.
- ❖ Implementation of an HMI supervision interface to manage the operation of the machines.

4. Conclusion

After presenting the host company, its activities and its different products. In the next chapter, we have moved on to the general study of our system, as well as the problems of our graduation project.

In the next chapter, we will introduce the Blow room line machines.

Chapter 2: Introduction to the machine

1. Introduction

Understanding and mastering how a system works in order to be able to automate it is an essential step. This is why we started our end-of-studies internship by studying the main components involved in the operation of the Blow room line machines .

2. Functional analysis of the Blow room line

The Blow room section plays a critical role in preparing raw cotton for spinning. It ensures that fibers are opened, cleaned, blended, and ready for subsequent processing steps.

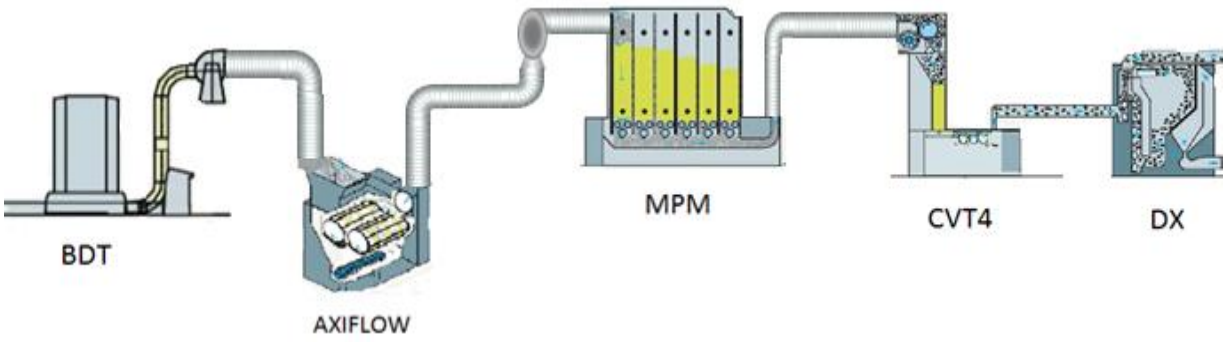


Figure 17 : Blow room line

2.1 BDT

2.1.1 Operation

The BDT is a bale opening machine which opens the compressed fibers into very small tufts to increase the volume and to create new surfaces to work with for further processing .
Motor M1 moves the BDT horizontally until the detection of cotton.
Motor M2 is attached with a carpet to cover the canal of the transferred cotton.
Motor M3 moves the BDT arm vertically until the detection of the cotton's level .
Motor M4 adjusts the positions of M5 and M6 .
Motor M5 and M6 are two motors working simultaneously to open the cotton into small tufts .
Motor M7 press on the bales to obtain uniform height.

2.1.2 Structure

This is the bale-opening machine BDT from the brand " **TRÜTZSCHLER"** as presented below:

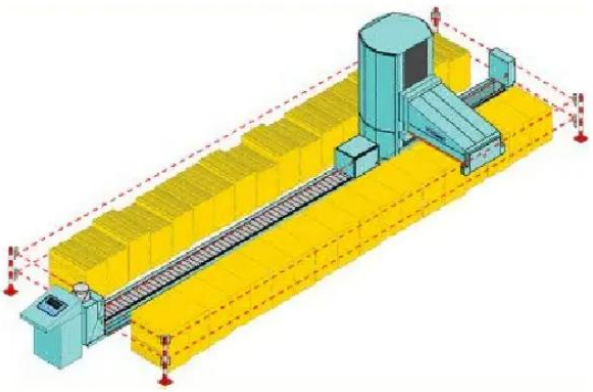


Figure 18 : BDT

2.1.3 Components

The figure below demonstrate the internal components of BDT:

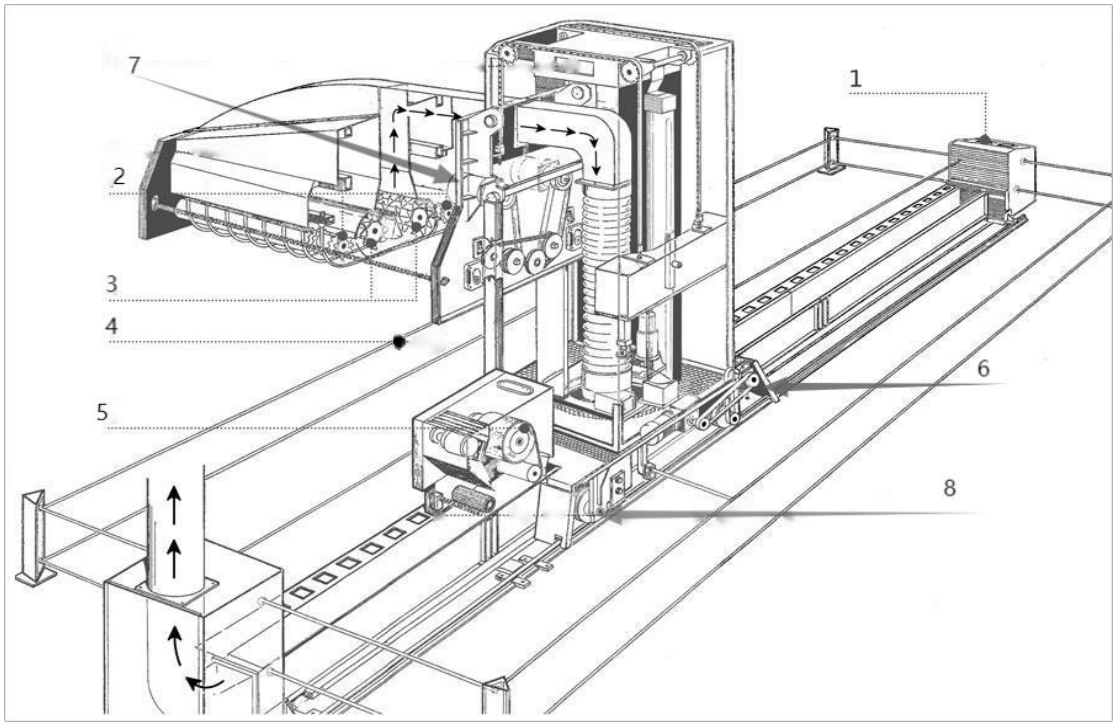


Figure 19: Internal structure

Table 2: BDT components

No	Name
1	Control Panel
2	Pressing Motor
3	Opening Motors
4	Photocell Safety Sensor
5	Carpet Motor
6	Movement blocking Sensor
7	Vertical Translation Motor
8	Horizontal Translation Motor

2.1.3.1 Sensors

Sensor SD is for cotton detection.

Sensor SH is for cotton's height detection.

Sensor SI detects the initial position of BDT

Sensor SF detects the final position of BDT

Sensor SMIN detects the lowest level that the arm could reach.

Sensor SMAX detects the highest level that the arm could reach.

2.1.3.2 Security

Sensor S1 is a movement blocking sensor .

Sensor S2 is a door safety sensor.

Sensor S3 is a photocell safety sensor to detect human presence .

Sensor S4 Flame sensor.

Emergency stop button in case of machine failure

2.2 Technical Fan 1

2.2.1 Operation

The Technical Fan acts as an intermediate between BDT and AXIFLOW transferring the opened fibers for the cleaning process.

Motor M1 converts electrical energy into mechanical energy, which is then used to rotate the fan blades and generate airflow .

2.2.2 Structure

This is the Technical Fan 1 as shown below :



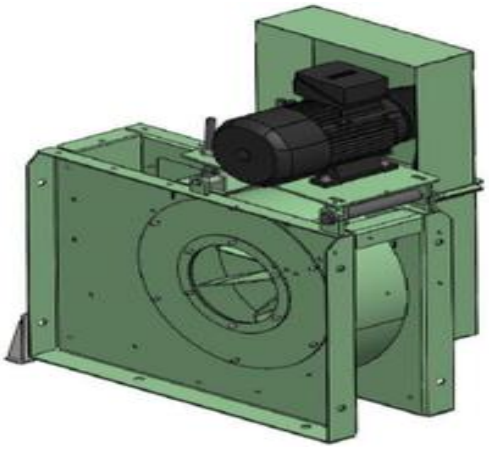


Figure 20: Technical Fan 1

2.2.3 Components

2.2.3.1 Motor

The figure below shows the motor 1 of Technical Fan 1 :

Figure 21: Motor M1

2.2.3.2 Security

Sensor S1 Flame sensor.

2.3 AXIFLOW

2.3.1 Operation

The AXIFLOW removes dust and impurities from cotton fibers by beating them with 2 rollers.

This mechanical action expel foreign particles , ensuring cleaner and more refined material .

Motor 1 drives the 2 rollers by rotational movement beating the cotton ensuring efficient cleaning .

With the gravity effect the impurities falls down into small bucket.

Motor 2 extract the fallen dust and impurities from the bucket .

2.3.2 Structure

This is the pre-cleaning machine AXIFLOW as shown below:

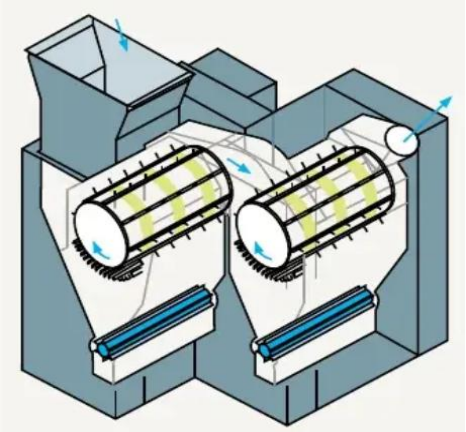


Figure 22: AXIFLOW

2.3.3 Components

2.3.3.1 Motors

The figures below represent the 2 motors of AXIFLOW

Figure 23: Motor 1

Figure 24: Motor 2

2.3.3.2 Security

-Sensor S1 Flame sensor.

-Sensor S2 is a door safety sensor.

Figure 25:Door safety sensor

2.4 Technical Fan 2

2.4.1 Operation

The Technical Fan acts as an intermediate between AXIFLOW and MPM transferring the opened fibers for mixing and blending process.

Motor M1 converts electrical energy into mechanical energy, which is then used to rotate the fan blades and generate airflow .

2.4.2 Structure

This is the Technical Fan 2 structure as shown below:

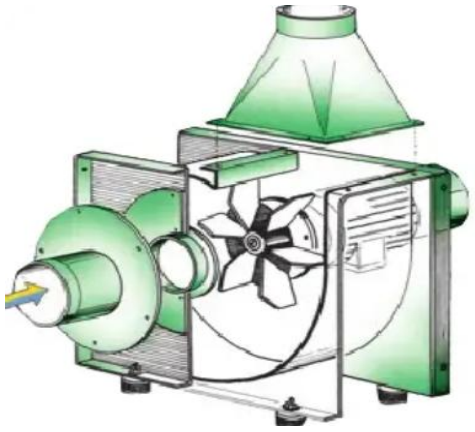


Figure 26:Technical fan 2

2.4.3 Components

2.4.3.1 Motor

The figure below shows the motor 1 of Technical Fan 2

Figure 27:Motor 1

2.4.3.2 Security

-Sensor S1 Flame sensor.

2.5 MPM

2.5.1 Operation

The MPM role is to achieve a uniform blending of cotton fibers from various sources. By ensuring consistent mixing.

The Technical fan 2 transfer the cotton in the canal to fill the 10 chambers.

The 2 Delivery Motors 1 for each 5 chambers and its used to deliver the cotton to the opening motors.

The 2 Opening Motors 1 for each 5 chambers and its used to open the cotton in the canal facilitating the emptying process .

Pressure Switch sensor operates as an electrical contact when a certain pressure reaches .

The 9 valves(flaps) are for sealing the chambers.

2.5.2 Structure

This is the multi-mixer machine MPM from the brand " **TRÜTZSCHLER" :**

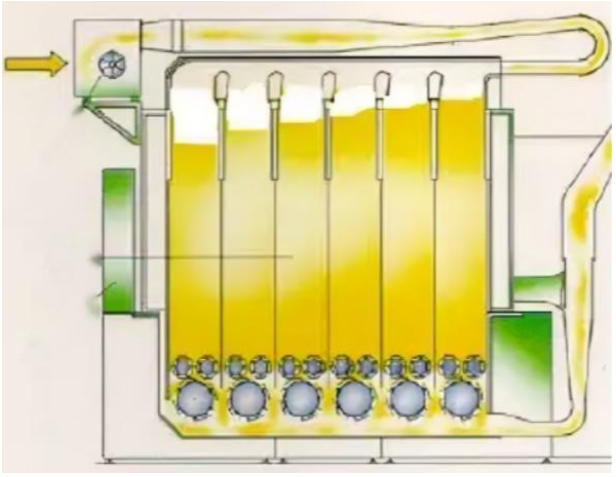


Figure 28: MPM

2.5.3 Components

2.5.3.1 Motors

The figures below shows the MPM motors() :

Figure 29: 2 Delivery motors

Figure 30: 2 Opening motors

2.5.3.2 Sensors

The pressure switch is a pressure sensor, which ensures the detection of pressure in the pipe that is managed by the fan. The adjustment of the air pressure to be detected is carried out manually. And depending on the value displayed by the pressure switch, the system is either up or down.



Figure 31:Pressure switch

2.5.3.3 Security

-Emergency stop button in case of machine failure

-Sensor S2 is a door safety sensor.

-Sensor S1 Fire sensor.

Figure 32: Fire sensor

2.6 LVSA Condenser

2.6.1 Operation

The LVSA Condenser contributes to efficient opening, cleaning, and blending with its two opening rolls working simultaneously .

Motor M1 transfer the cotton to the motor 2.

Motor M2 drops down the cotton and fills the chamber .

2.6.2 Structure

This is the LVSA Condenser as represented below:

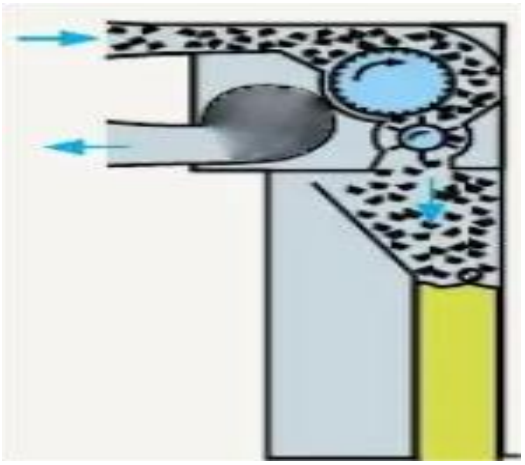


Figure 33: LVSA Condenser

2.6.3 Components

2.6.3.1 Motors

The figures shows the 2 motors(M1, M2) of LVSA Condenser:

Figure 34:Motor 1

Figure 35:Motor 2

2.6.3.2 Sensors

Sensor SCMAX Maximum level Sensor indicates that the chamber is full

Sensor SCMIN Maximum level Sensor indicates that the chamber is low

2.6.3.3 Security

-Sensor S4 Flame sensor.

-Sensor S2 is a door safety sensor.

2.7 CVT4

2.6.1 Operation

The CVT4 is a fine-cleaning machine , the cotton is obtained from LVSA condenser with a roller carpet that transfer the cotton to the 4 cylinders , these cylinders perform a profound beating action to

disentangle the fibers and separates them from each other, ensuring a high level of cleanliness.

-Motor MC rolls the carpet transferring the cotton to the 4 cylinders.

-The 4 Motors rotates the cylinders to perform the beating action.

-4 speed sensors to activate the cylinders right after the wanted speed is reached.

2.6.2 Structure

This is the fine cleaning machine CVT4 from the brand " **TRÜTZSCHLER" :**

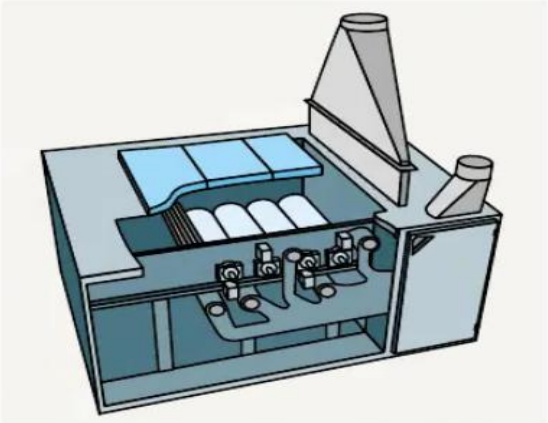


Figure 36: CVT4

2.6.3 Components

2.6.3.1 Motors

The figure below represents the main motors(MC, M1, M2, M3, M4) of CVT4 :

Figure 37: CVT4 motors

2.6.3.3 Sensors

-S1 speed sensor for the 1st cylinder.

-S2 speed sensor for the 2nd cylinder.

-S3 speed sensor for the 3rd cylinder.

-S4 speed sensor for the 4th cylinder.

2.6.3.1 Security

-Emergency stop button in case of machine failure

-Sensor S1 Flame sensor.

-Sensor S2 is a door safety sensor.

2.8 DX

2.8.1 Operation

The DX is the final machine in Blow room, its primarily function is to remove the dust .

Motor M1 receive the cotton from CVT4 and transfer to DX.

Motor M2 spread the cotton over the machine to clean the dust.

Motor M3 transfer the cotton from DX and transfer to the Card.

2.8.2 Structure

This is the dust cleaner machine DX as shown below :

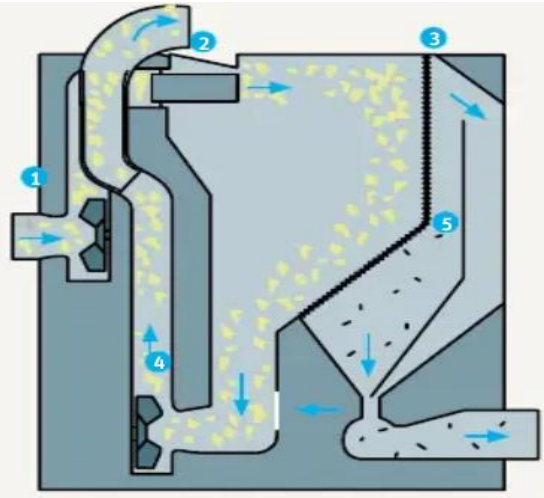


Figure 38 : DX

2.8.3 Components

2.8.3.1 Motors

The figures below represents the 3 motors(M1, M2, M3) of DX :

Figure 39: Motor 1
Figure 40: Motor 2

Figure 41: Motor 3

2.8.3.2 Security

-Emergency stop button in case of machine failure

-Sensor S1 Flame sensor.

-Sensor S2 is a door safety sensor.

Figure 42: Door safety sensor

3. Storage Unit

The storage unit serves as the central point of control, enabling the operation of various machines' components.



Figure 43: Storage Unit

3.1 Electrical cabinet

The electrical cabinet contains the main components of the system circuit.

3.1.1 Structure



Figure 44: Internal structure

3.1.2 Components

3.1.2.1 Circuit Breaker

A circuit breaker is a protective device whose function is to interrupt the electric current in the event of an overload or short circuit.



Figure 45: Circuit Breaker

3.1.2.2 Contactor

The contactor is a mechanical connection device with a single resting position and a single working position. It is capable of establishing, withstanding, and interrupting currents under normal circuit conditions, including in-service overload conditions. Three-pole contactors usually have an auxiliary contact.



Figure 46: Contactor

3.1.2.2 Transformer

The transformer transfers electrical energy from one circuit to another using electromagnetic induction.

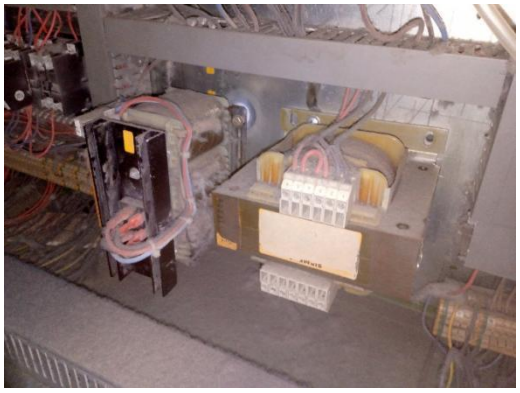


Figure 47: Transformer

4. Conclusion

In this chapter I have made a definite study of the system and its current operation and features and then proposed a suitable modification for its automation.

I'm going to deal with the different GRAFCETs of the system in the next chapter.

Chapter 3: Grafcet

1. Introduction

In this chapter, we focus on the study of the operation of the Blow room line machines through the development of the appropriate GRAFCETs and then the realization of the different wiring of the system.

2. How the system works

2.1 Defects to be detected

There are many defects that can be detected:

- If the door safety of BDT is opened the motors M1 M2 M3 M4 M5 M6 and M7 stops.
- If the door safety of AXIFLOW is opened the motors M1AX and M2AX stops.
- If the door safety of MPM is opened the motors MW1 MW2 MF1 MF2 stops.
- If the door safety of LVSA is opened the motors MC1 and MC2 stops.
- If the door safety of CVT4 is opened the motors MC and the 4 motors stops.
- If the door safety of DX is opened the motors M1DX and M2DX stops.
- If the fire sensor of any machine detects fire the BDT Axiflow MPM LVSA CVT4 and DX stops.
- If the photocell detector of BDT detects human presence the motors M1 M2 M3 M4 M5 M6 and M7 stops.
- If the movement blocking sensor is triggered the motors M1 M2 M3 M4 M5 M6 and M7 stops.

2.2 Proposed automatic operating principle

2.2.1 BDT

✓ The target zone is manually filled by the user, then the user press on start button .
✓ The motors M1, M2, M3, M4 begin operating , then the sensor SC detects cotton.
✓ The motor M descent until it reaches the sensor height of cotton .
✓ The motors M1, M2, M3, M4 begin operating, until there is no cotton detection .
✓ Then the cycle repeat itself until the minimum arm height sensor is reached .
✓ Then the machine returns back to its initial position.

2.2.2 MPM

✓ When the user press on start button and selects AUTO mode the MPM starts filling and emptying the chambers simultaneously .
✓ If the maximum sensor of the condenser is on then the MPM stops emptying .
✓ If the minimum sensor of the condenser is on then the MPM starts emptying again.
✓ When the chamber of LVSA is fully filled then the MPM stops filling.

2.2.3 CVT4

✓ The users press on start button .

✓ The 4 cylinders start rotating starting from the 4th to the 1st .

✓ The carpet starts transferring the cotton to the cylinders after the required speed of the 1st motor is reached.

2.3 Automatic operation process

2.3.1 Mode Auto

- In the automatic operating mode, the user press on Start global button and each machine start as they work continuously without human interference .
- The BDT start opening bales and then the Technical Fan transfer the opened fibers to AXIFLOW.
- The AXIFLOW start cleaning the cotton with its beater rollers.
- The MPM receive the pre-cleaned cotton with its Technical Fan and then storing and mixing .
- The LVSA condenser receive the mixed cotton from MPM and transfer it to CVT4 .
- The CVT4 start cleaning the cotton and disentangle the fibers from each others.
- The DX remove the dust.

2.3.2 Security part

During production, the machines detects any defects that may occur and which are all signaled by the same signal lamp. The defects that can be detected are as follows:

- If a fire sensor detects fire all machines of the Blow room line stops .
- If the Movement Blocking sensor of BDT has been triggered all Motors of the BDT stops.
- If the Photocell safety sensor of BDT detects a human presence all Motors of the BDT stops.
- If one of the doors of the Blow room line machines opens all motors of that machine stops.

After any fault the machine resumes normal operation after the fault has been repaired .

3. The Grafcets

3.1 Definition of Grafcet

GRAF is the abbreviation of the expression **"Functional Graph for the Control of Stages and Transitions"** is a tool that allows you to graphically and structurally represent the operation of sequential automation using step and transition.

3.2 Basic Elements of Grafcet

This Grafcet is a series of transitions and includes:

- ✓ Steps with actions associated with them.
- ✓ Transitions to which receptivities are associated.
- ✓ Oriented links connecting steps to transitions and transitions to steps.

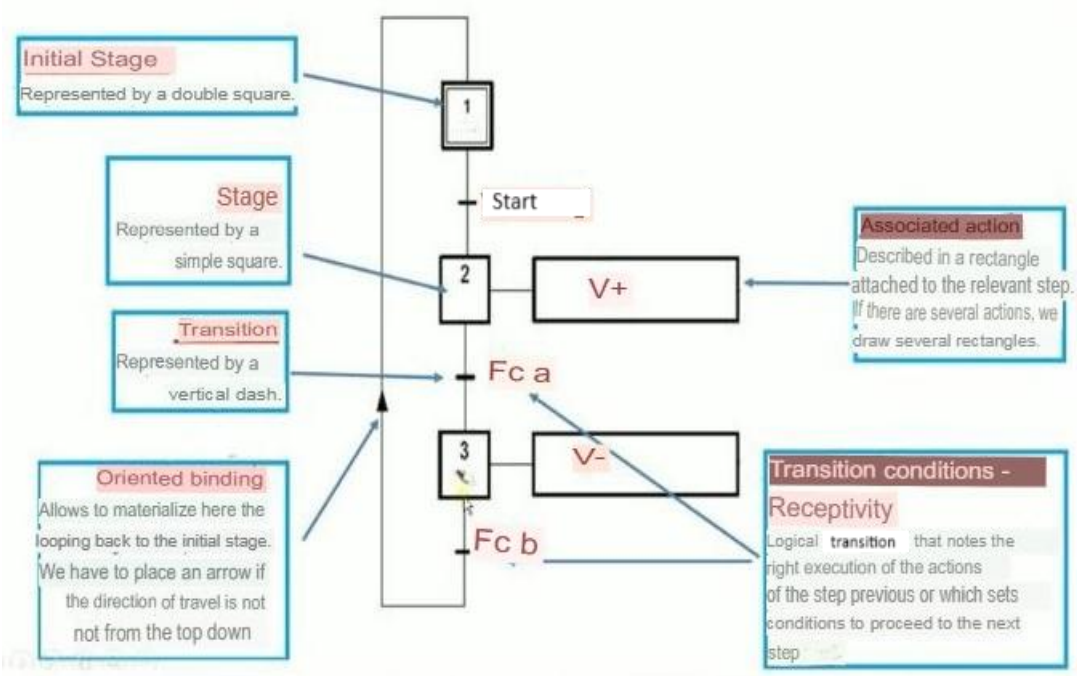


Figure 48: Basic elements of Grafcet

4. Implementation

Our specifications include a part of the development of the program with the software **TIA PORTAL.**

To do this, we proceeded with the development of the **GRAFCET** of our system

4.1 Viewpoint Grafcet Command

This GRAFCET is established by specifying the technology of the following dialog elements:

- Between the Ordering Part and the Operational Part.
- Between Control Party and Operator.
- Between Part Control and other system.

It is a GRAFCET drawn up by a specialist, it is the version that allows him to establish the equations and possibly the implementation diagrams (electrical, pneumatic, etc.)

4.1.1 Grafcet BDT

This GRAFCET is responsible for the start-up of BDT and the operation of the main motors .

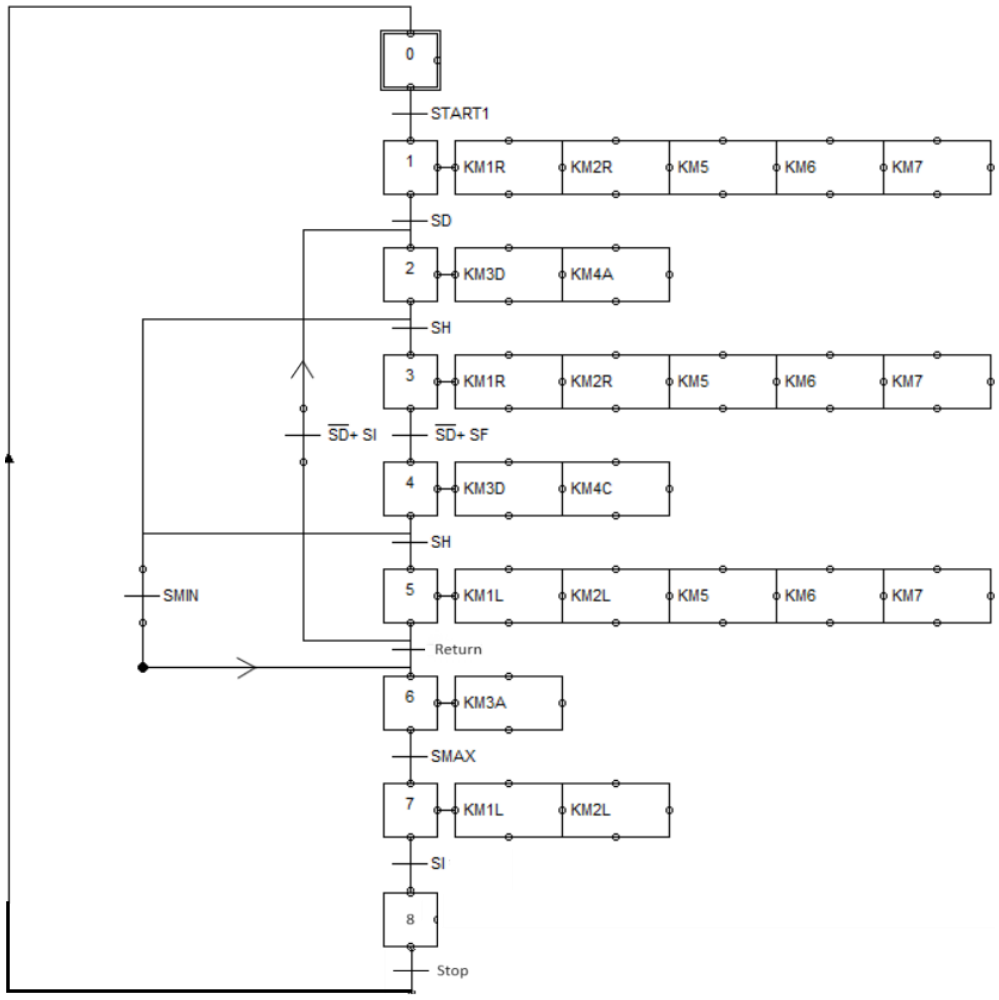


Figure 49: Grafcet BDT

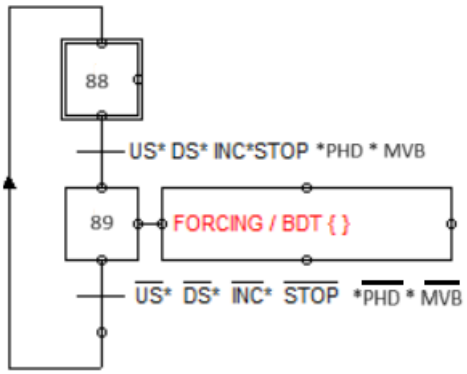


Figure 50:Security Grafcet

Tableau 3: BDT elements

	Name	Action	Symbol
INPUT	Start button	Start BDT	START1
	Cotton detection sensor	Detect cotton	SD
	Cotton's height sensor	Detect cotton's height	SH
	Sensor of final position	Detect final position	SF
	Sensor of initial position	Detect initial position	SI
	Minimum arm level sensor	Detect minimum arm level	SMIN
	Maximum arm level sensor	Detect maximum arm level	SMAX
	Stop button	Stop BDT	STOP1
	Return button	Return machine	Return
	Emergency stop button	Urgent stop	US
	Door safety sensor	Detect if someone opens the machine	DS
	Fire sensor	Detect fire	INC
	Photocell sensor	Detect human presence	PHD
	Movement block sensor	Detect if the movement is blocked	MVB
OUTPUT	Motor 1 right direction		KM1R
	Motor 1 left direction		KM1L

	Carpet Motor 2 right direction		KM2R
	Carpet Motor 2 left direction		KM2L
	Arm motor 3 downward		KM3D
	Arm motor 3 upward		KM3A
	Motor 4 anti-clockwise rotation		KM4A
	Motor 4 clockwise rotation		KM4C
	Opening fibers motor 5		KM5
	Opening fibers motor 6		KM6
	Pressing motor 7		KM7
Step	Initial Step		X0
	Step 1	Moving right to detect cotton	X1
	Step 2	Adjust arm level to match the cotton's height	X2
	Step 3	Start opening fibers while moving right	X3
	Step 4	Adjust arm level to match the cotton's height	X4
	Step 5	Start opening fibers while moving left	X5
	Step 6	Ascent arm to the maximum level	X6
	Step 7	Back to the initial position	X7
	Step 8	Waiting	X8

4.1.2 Grafcet Technical Fan 1

This Grafcet ensures the start-up of the Technical fan 1.

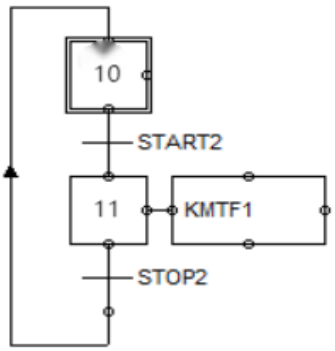


Figure 51: Grafcet Technical fan 1

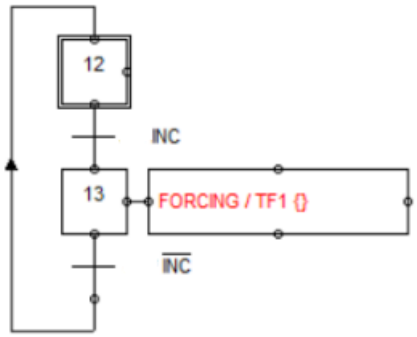


Figure 52: Security Grafcet

Tableau 4: Technical fan 1 elements

	Name	Action	Symbol
INPUT	Start button	Start technical fan	Start2
	Stop button	Stop technical fan	Stop2
	Fire sensor	Detect fire	INC
OU	Motor technical fan 1		KMTF1
Step	Initial step		X10
	Step 11	Receive and transfer cotton	X11
	Initial step		X12
	Step 13	Forcing reset	X13

4.1.3 Grafcet AXIFLOW

This GRAFCET is responsible for the start-up of Axiflow and the operation of the main motors .

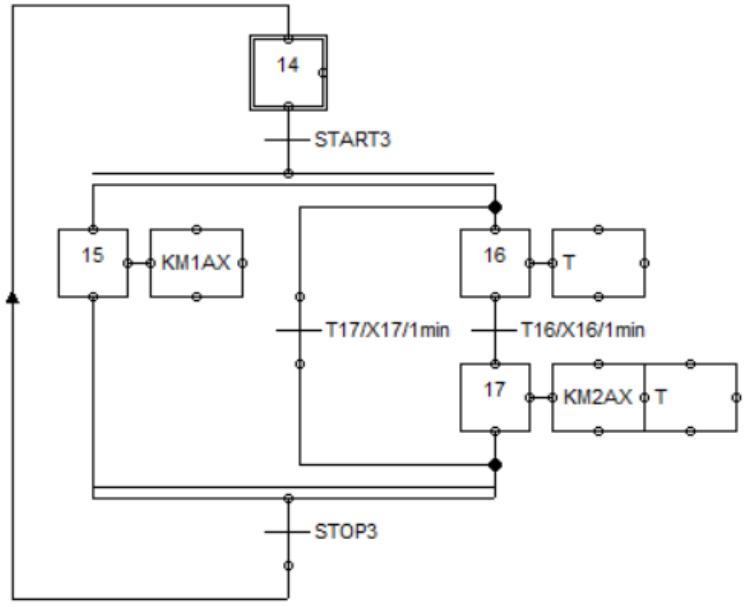


Figure 53: Grafcet Axiflow

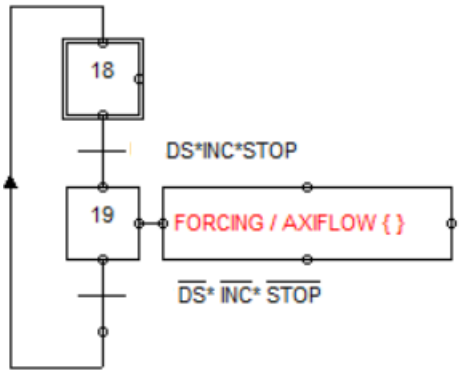


Figure 54: Security Grafcet

Tableau 5: Axiflow elements

	Name	Action	Symbol
INPUT	Start button	Start Axiflow	Start3
	Stop button	Stop Axiflow	Stop3
	Door safety sensor	Detect if someone opens the machine	DS
	Fire sensor	Detect fire	INC

	Delay	Step 16 delay for 1min	T16/X16/ 1min
	Delay	Step 17 delay for 1min	T17/X17/ 1min
OUTPUT	Motor 1 AX	Cleaning cotton	KM1AX
	Motor 2 AX	Extracting dust	KM2AX
	Delay	Waiting step	T
Step	Initial step		X14
	Step 15	Cleaning cotton	X15
	Step 16	Waiting step	X16
	Step 17	Extracting dust	X17
	Initial step		X18
	Step 19	Forcing reset	X19

4.1.4 Grafcet Technical Fan 2

This Grafcet ensures the start-up of the Technical fan 2.

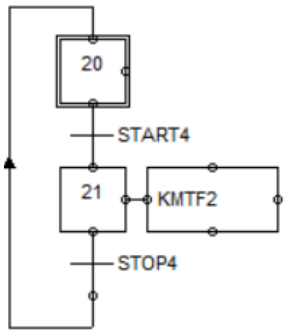


Figure 55: Grafcet Technical fan 2

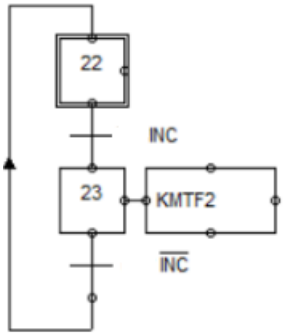


Figure 56: Security Grafcet

Tableau 6: Technical fan 2 elements

	Name	Action	Symbol
INPUT	Start button	Start technical fan 2	Start4
	Stop button	Stop technical fan 2	Stop4
	Fire sensor	Detect fire	INC
OU	Motor technical fan 2		KMTF2
Step	Step 10	Initial step	X20
	Step 11	Receive and transfer cotton	X21
	Step 22	Initial step	X22
	Step 23	Forcing reset	X23

4.1.5 Grafcet MPM

This GRAFCET ensure the startup of the MPM and there is 3 choices to choose between emptying , filling or the automatic options.

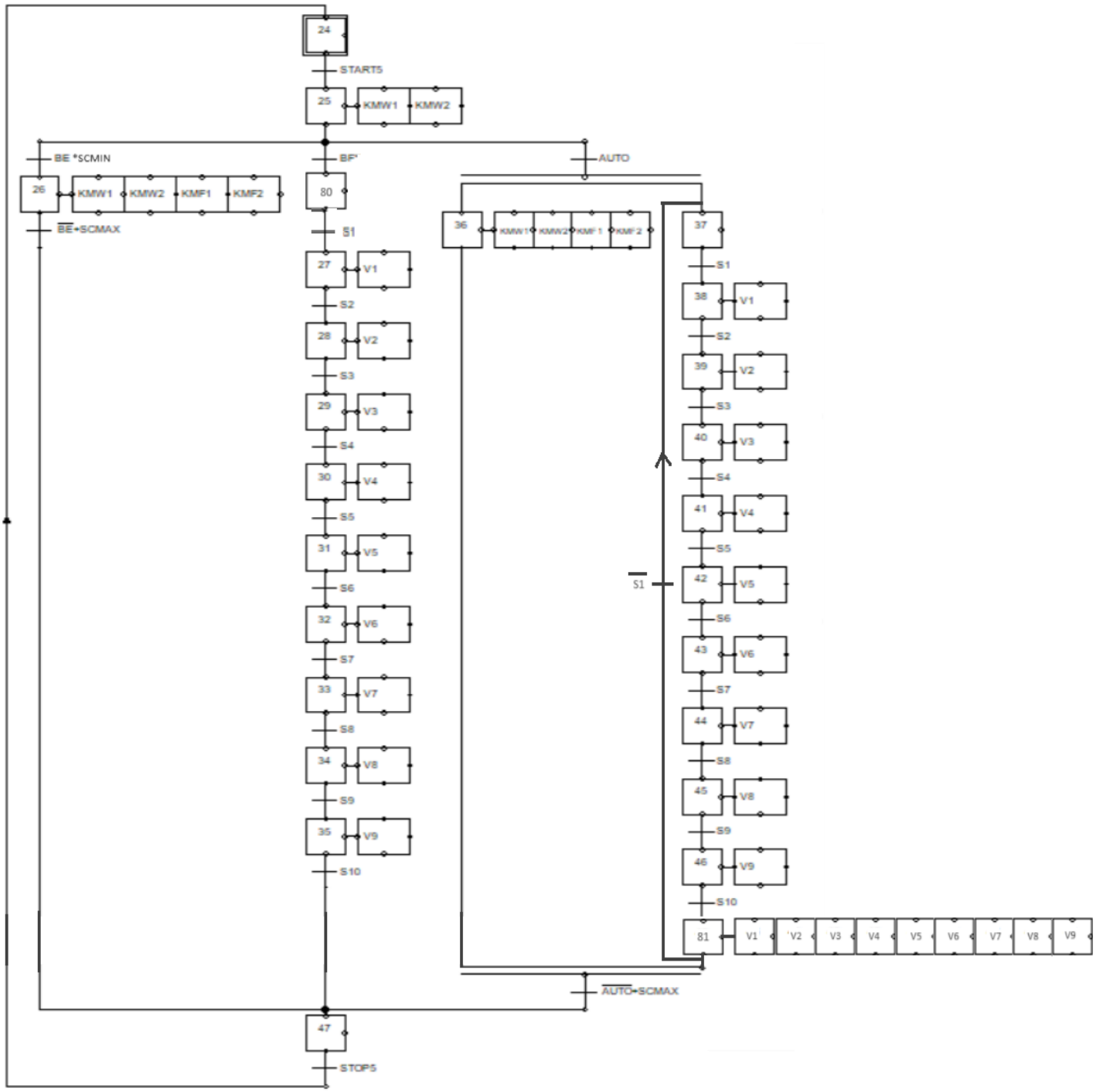


Figure 57: Grafcet MPM

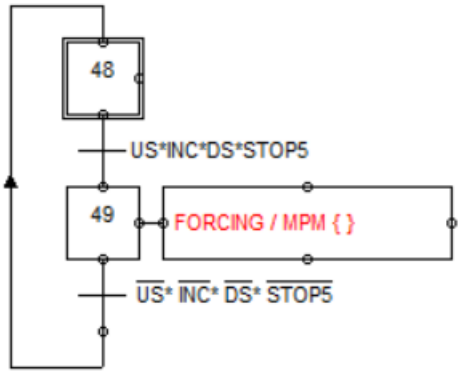


Figure 58: Security Grafcet

Tableau 7: MPM elements

	Name	Action	Symbol
INPUT	Start button	Start MPM	Start5
	Stop button	Stop MPM	Stop5
	Emergency stop button	Urgent stop	US
	Door safety sensor	Detect if someone opens the machine	DS
	Fire sensor	Detect fire	INC
	Emptying button	Start emptying	BE
	Filling button	Start filling	BF
	Automatic button	Start filling and emptying automatically	AUTO
	Condenser maximum level sensor	Detect if condenser is full	SCMAX
	Condenser minimum level sensor	Detect if condenser is low	SCMIN
	Chamber 1 maximum level sensor	Detect if chamber 1 is full	S1
	Chamber 2 maximum level sensor	Detect if chamber 2 is full	S2
	Chamber 3 maximum level sensor	Detect if chamber 3 is full	S3
	Chamber 4 maximum level sensor	Detect if chamber 4 is full	S4
	Chamber 5 maximum level sensor	Detect if chamber 5 is full	S5
	Chamber 6 maximum level sensor	Detect if chamber 6 is full	S6
	Chamber 7 maximum level sensor	Detect if chamber 7 is full	S7
	Chamber 8 maximum level sensor	Detect if chamber 8 is full	S8
	Chamber 9 maximum level sensor	Detect if chamber 9 is full	S9

	Chamber 10 maximum level sensor	Detect if chamber 10 is full	S10
OUTPUT	Opening motor 1		KMW1
	Opening motor 2		KMW2
	Delivery motor 1		KMF1
	Delivery motor 2		KMF2
	Flap 1	Sealing chamber 1	V1
	Flap 2	Sealing chamber 2	V2
	Flap 3	Sealing chamber 3	V3
	Flap 4	Sealing chamber 4	V4
	Flap 5	Sealing chamber 5	V5
	Flap 6	Sealing chamber 6	V6
	Flap 7	Sealing chamber 7	V7
	Flap 8	Sealing chamber 8	V8
	Flap 9	Sealing chamber 9	V9
Step	Step 24	Initial step	X24
	Step 25	Emptying canal	X25
	Step 26	Emptying chambers and canal	X26
	Step 27	Sealing chamber 1	X27
	Step 28	Sealing chamber 2	X28
	Step 29	Sealing chamber 3	X29
	Step 30	Sealing chamber 4	X30
	Step 31	Sealing chamber 5	X31
	Step 32	Sealing chamber 6	X32
	Step 33	Sealing chamber 7	X33
	Step 34	Sealing chamber 8	X34
	Step 35	Sealing chamber 9	X35
	Step 36	Emptying chambers and canal automatically	X36
	Step 37	Waiting	X37
	Step 38	Sealing chamber 1	X38
	Step 39	Sealing chamber 2	X39

	Step 40	Sealing chamber 3	X40
	Step 41	Sealing chamber 4	X41
	Step 42	Sealing chamber 5	X42
	Step 43	Sealing chamber 6	X43
	Step 44	Sealing chamber 7	X44
	Step 45	Sealing chamber 8	X45
	Step 46	Sealing chamber 9	X46
	Step 47	Waiting	X47
	Step 48	Initial step	X48
	Step 49	Forcing reset	X49
	Step 80	Waiting	X80
	Step 81	Reset	X81

4.1.6 Grafcet LVSA Condenser

This Grafcet ensures the start-up of the LVSA Condenser and startup delay between the 2 motors .

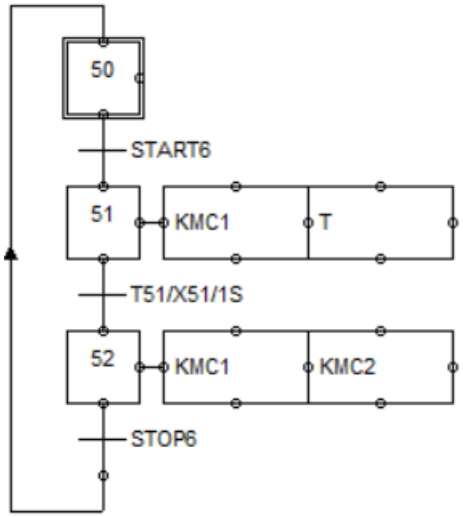


Figure 59: Grafcet LVSA Condenser

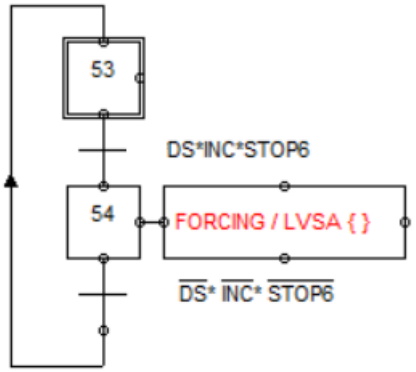


Figure 60:Security Grafcet

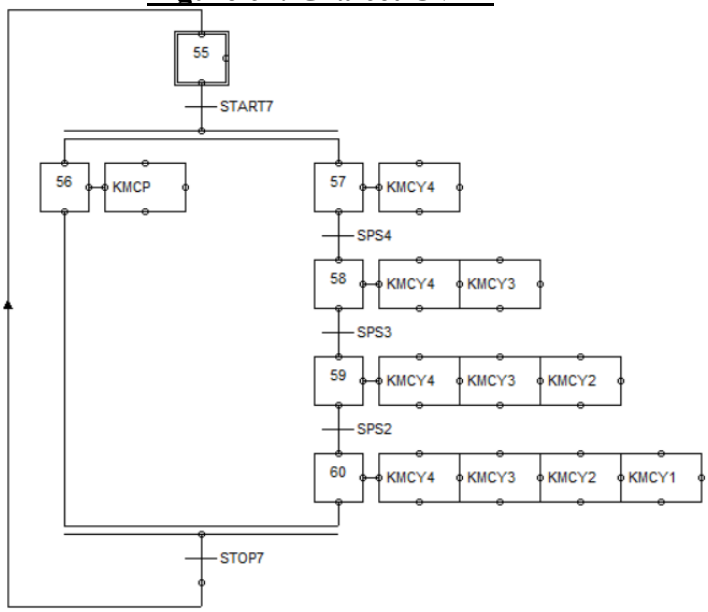
Tableau 8: LVSA elements

	Name	Action	Symbol
INPUT	Start button	Start LVSA	Start6
	Stop button	Stop LVSA	Stop6
	Door safety sensor	Detect if someone opens the machine	DS
	Fire sensor	Detect fire	INC
	Delay	Step 51 delay for 1s	T51/X51/1S
OUTPUT	Motor 1 of condenser		KMC1
	Motor 2 of condenser		KMC2
Step	Step 50	Initial step	X50
	Step 51	Receive and transfer cotton	X51
	Step 52	Fill the condenser chamber	X52
	Step 53	Initial step	X53
	Step 54	Forcing reset	X54

4.1.7 Grafcet CVT4

This GRAFCET ensures the startup of CVT4 and the parallel operation between the carpet motor and the 4 motors.

Figure 61: Grafcet CVT4



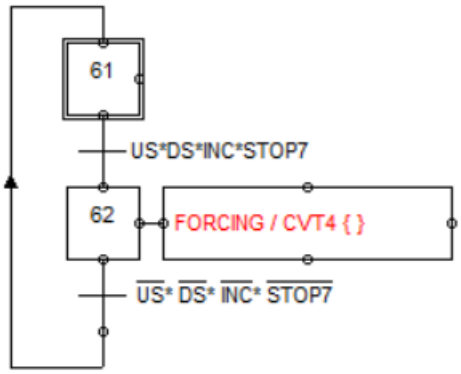


Figure 62: Security Grafcet

Tableau 9: CVT4 elements

	Name	Action	Symbol
INPUT	Start button	Start CVT4	Start7
	Stop button	Stop CVT4	Stop7
	Emergency stop button	Urgent stop	US
	Door safety sensor	Detect if someone opens the machine	DS
	Fire sensor	Detect fire	INC
	Speed sensor	Detect the speed limit of cylinder4	SPS4
	Speed sensor	Detect the speed limit of cylinder3	SPS3
	Speed sensor	Detect the speed limit of cylinder2	SPS2

OUTPUT	Carpet Motor		KMCP
	Motor cylinder 4		KMCY4
	Motor cylinder 3		KMCY3
	Motor cylinder 2		KMCY2
	Motor cylinder 1		KMCY1
Step	Step 55	Initial step	X55
	Step 56	Transfer cotton to the cylinders	X56
	Step 57	Starting motor 4 until speed limit reaches	X57
	Step 58	Starting motor 4 and 3 until speed limit reaches	X58
	Step 59	Starting motor 4, 3 and 2 until speed limit reaches	X59
	Step 60	Clean and disentangle the cotton	X60
	Step 61	Initial step	X61
	Step 62	Forcing reset	X62

4.1.8 Grafcet DX

This Grafcet ensures the start-up of the DX and startup delay between the 3 motors .

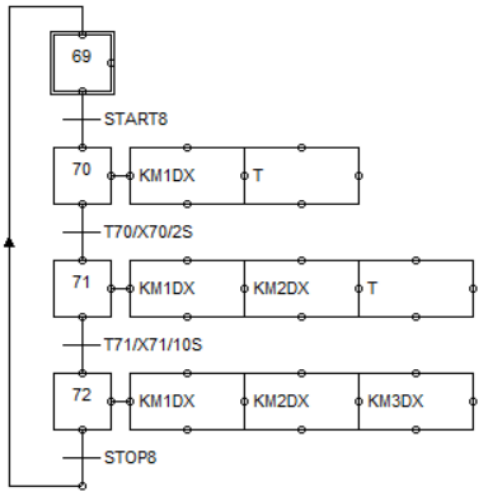


Figure 63: Grafcet DX

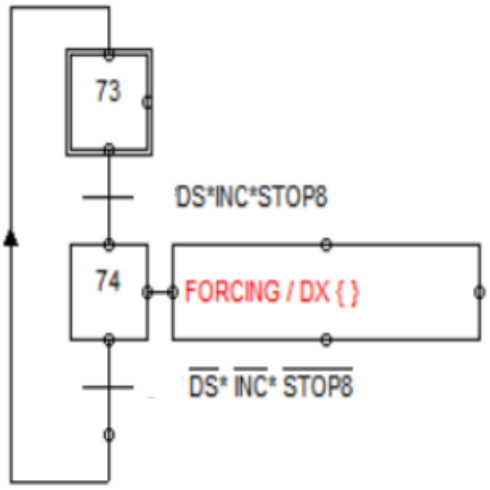


Figure 64: Security Grafcet

Tableau 10 : DX elements

	Name	Action	Symbol
INPUT	Start button	Start DX	Start8
	Stop button	Stop DX	Stop8
	Door safety sensor	Detect if someone opens the machine	DS
	Fire sensor	Detect fire	INC
	Delay	Step 70 delay for 2s	T70/X70/ 2S
	Delay	Step 51 delay for 10s	T70/X70/ 10S
OUTPUT	Motor 1 DX		KM1DX
	Motor 2 DX		KM2DX
	Motor 3 DX		KM3DX
Step	Step 69	Initial step	X69
	Step 70	Receiving cotton	X70
	Step 71	Receiving and Spreading cotton	X71
	Step 72	Receiving, Spreading and transferring cotton	X72
	Step 73	Initial step	X73
	Step 74	Forcing reset	X74

5. Conclusion

In this chapter we are interested in the presentation of the **GRAFCET** of the system and its translation into contact language to finally do the simulation.

In the next chapter, we will present the automation and supervision tasks of the Blow room machines.

Chapter 4: Automation and supervision

1. Introduction

In this chapter, we move on to the practical application of the operation of the new system using **TIAPORTALV17** and its tools for the programming of the programmable logic controller, the simulation of the program and the realization of the HMI.

2. Description of Enhancements

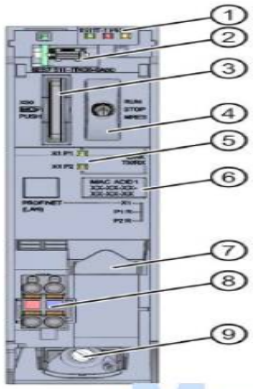
A few changes have been made, which are as follows:

- I have changed most of the commands that are necessary for the system to work automatically.
- I created an "**HMI**" communication interface between the user and the PLC to replace the old cabled system.

3. Automation via PLC and HMI

3.1 Definition of a Programmable Logic Controller

A Programmable Logic Controller (**PLC**) is an electronic machine that can be programmed by non-computer scientists and is designed to control industrial processes in an industrial environment and in real time. A programmable logic controller (PLC) can be adapted to a maximum of applications, from a processing, component and language point of view. That's why it's modular.



ManualsLib.com

1. LEDs for the current operating mode and diagnostics status of the CPU
2. Display connection
3. Slot for the SIMATIC memory card
4. Mode selector
5. LEDs for the 2 ports of the PROFINET interface X1
6. MAC address
7. PROFINET IO interface (X1) with 2 ports
8. Connection for supply voltage
9. Fastening screw

Figure 65: SIEMENS Programmable Logic Controller

All PLCs operate according to the same procedure:

- ✓ **In-house processing**

The PLC carries out control operations and updates certain system parameters (detection of **RUN/STOP** passages, timestamp desalination updates, etc.)

- ✓ **Reading Inputs**

The PLC reads the inputs (synchronously) and copies them into the image memory of the inputs.

- ✓ **Program Delivery**

The PLC executes the program instruction by instruction and writes the outputs to the output image memory.

- ✓ **Writing Outputs**

The PLC switches the different outputs (synchronously) to the positions defined in the image memory of the outputs

These four operations are carried out continuously by the PLC

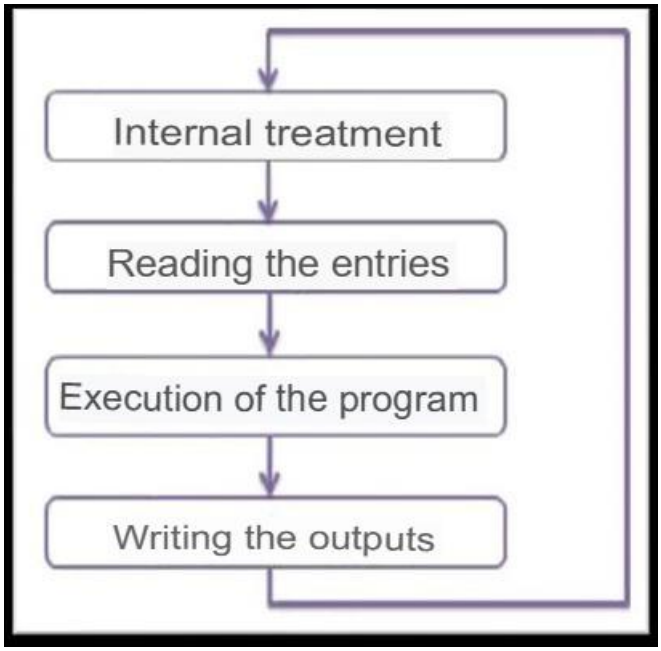


Figure 66: How a PLC Works

3.2 Choosing the PLC

In our project, we chose the "**SIMATIC S7-1500" Industrial Programmable Logic Controller** This choice is made after studying the machine and identifying the inputs and outputs of our machine and executing instructions at high speed. It is suitable for both simple industrial automation tasks and complex automation tasks. The range is available in several CPU types (1511-1513) in the DC24V power supply version.



Figure 67: PLC S7 -1500

3.2.1 Introducing the SIMATIC S7 1500

The **"S7-1500" family** consists of programmable logic controllers (**PLCs**) that can be used in a variety of automation applications. Its compact design, low price and large instruction set make it an ideal solution for controlling a wide variety of applications. The **"S7-1500"** models and **Windows-based** programming tool give you the flexibility to solve your automation problems. This manual, which provides information on the installation and programming of the S7-1500 PLCs, is designed for engineers, programmers, installers, and electricians with a general knowledge of programmable logic controllers.

3.2.2 Description of the PLC

- 32 Digital INPUT 24V DC
- 32 Digital OUTPUT 24V DC
- 2 Analog OUTPUT 24V DC
- 5 Analog INPUT 24V DC
- Work memory 250 KB
- frequency output (up to 100kHz)
- Interface profit
- HMI and PLC-to-PLC **communication**
- Powering DC 24V

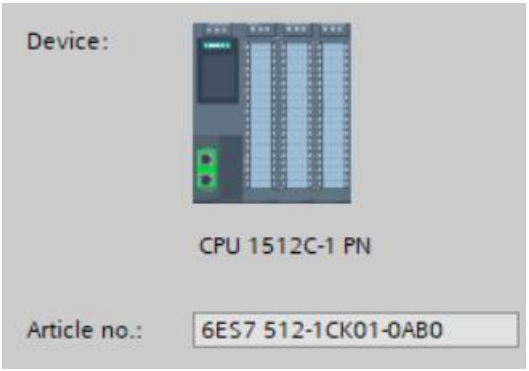


Figure 68: Automate CPU 1512C-1 PN

3.2.3 Choice of I/O Module

3.2.3.1 Digital Input Module (DI)

Name	Tag table	Data type	Address
ES BDT	BDT	Bool	%I10.0
ES MPM	MPM	Bool	%I10.1
ES CVT4	CVT4	Bool	%I10.2
DS AXF(1)	AXF	Bool	%I10.3
DS BDT(1)	BDT	Bool	%I10.4
SMAX(1)	BDT	Bool	%I10.5
SF(1)	BDT	Bool	%I10.6
SI(1)	BDT	Bool	%I10.7
SH(1)	BDT	Bool	%I11.0
SD(1)	BDT	Bool	%I11.1
SMIN(1)	BDT	Bool	%I11.2
DS CVT4(1)	CVT4	Bool	%I11.3
DS DX(1)	DX	Bool	%I11.4
SCMAX(1)	LVSA	Bool	%I11.5
S9(1)	MPM	Bool	%I11.6
S8(1)	MPM	Bool	%I11.7
S7(1)	MPM	Bool	%I12.0
S6(1)	MPM	Bool	%I12.1
S10(1)	MPM	Bool	%I12.2
S4(1)	MPM	Bool	%I12.3
S5(1)	MPM	Bool	%I12.4
DS MPM(1)	MPM	Bool	%I12.5
S2(1)	MPM	Bool	%I12.6
S3(1)	MPM	Bool	%I12.7
S1(1)	MPM	Bool	%I13.0
PHD BDT(1)	BDT	Bool	%I13.1
MVB BDT(1)	BDT	Bool	%I13.2
FS BDT(1)	BDT	Bool	%I13.3
FS TF1(1)	TF1	Bool	%I13.4
FS TF2(1)	TF2	Bool	%I13.5
FS AXF(1)	AXF	Bool	%I13.6
FS MPM(1)	MPM	Bool	%I13.7
FS LVSA(1)	LVSA	Bool	%I128.0
FS CVT4(1)	CVT4	Bool	%I128.1
FS DX(1)	DX	Bool	%I128.2
SPS4(1)	CVT4	Bool	%I128.3
SPS3(1)	CVT4	Bool	%I128.4
SPS2(1)	CVT4	Bool	%I128.5
DS LVSA(1)	LVSA	Bool	%I128.6

Figure 69: List of Digital Inputs

From the list you can see that the number of digital inputs is 39 So you can use:

- DI 16x24VDC BA
- 16 DI of Module

The module chosen is "6ES7 521-1BH10-0AA0"

3.2.3.2 Analog Input (AI) Module

Name	Tag table	Data type	Address
AI M CVT4	CVT4	Int	%IW0
AI D CVT4	CVT4	Int	%IW2
AI P MPM	MPM	Int	%IW4

Figure 70: Analog Input

For our machine, we mainly need 3 analog inputs for the pressure sensor.
For this, the 5 AI inputs of the S7-1512C-1 PN are enough .

3.2.3.3 Digital Output (DQ) Module

Name	Tag table	Data type	Address
KM2R	BDT	Bool	%Q4.0
KM1R	BDT	Bool	%Q4.1
KM3D	BDT	Bool	%Q4.2
KM5	BDT	Bool	%Q4.3
KM6	BDT	Bool	%Q4.4
KM7	BDT	Bool	%Q4.5
KM4A	BDT	Bool	%Q4.6
KM1L	BDT	Bool	%Q4.7
KM2L	BDT	Bool	%Q5.0
KM4C	BDT	Bool	%Q5.1
KMTF1	TF1	Bool	%Q5.2
KMTF2	TF2	Bool	%Q5.3
KM2AX	AXF	Bool	%Q5.4
KM1AX	AXF	Bool	%Q5.5
valve 8	MPM	Bool	%Q5.6
KMW1	MPM	Bool	%Q5.7
KMW2	MPM	Bool	%Q6.0
KMF1	MPM	Bool	%Q6.1
KMF2	MPM	Bool	%Q6.2
valve 1	MPM	Bool	%Q6.3
valve 2	MPM	Bool	%Q6.4
valve 3	MPM	Bool	%Q6.5
valve 4	MPM	Bool	%Q6.6
valve 5	MPM	Bool	%Q6.7
valve 6	MPM	Bool	%Q7.0
valve 7	MPM	Bool	%Q7.1
valve 9	MPM	Bool	%Q7.2
KMC1	LVSA	Bool	%Q7.3
KMC2	LVSA	Bool	%Q7.4
KMCY4	CVT4	Bool	%Q7.5
KMCP	CVT4	Bool	%Q7.6
KMCY1	CVT4	Bool	%Q7.7
KMCY2	CVT4	Bool	%Q128.0
KMCY3	CVT4	Bool	%Q128.1
KM3A	BDT	Bool	%Q128.2
KM1DX	DX	Bool	%Q128.3
KM2DX	DX	Bool	%Q128.4
KM3DX	DX	Bool	%Q128.5

Figure 71: Digital Output

From the list you can see that the number of digital outputs is 38. So you can use:

- DQ 8x24VDC/2A
- 8 DO of module

The DO Module used is "6ES7 522-1BF00-0AB0"

3.2.3.4 Analog Output (AO) Module

	AO DX	DX	Int	%QW0
	AO CVT4	CVT4	Int	%QW2

Figure 72: Analog Outputs

For our machines, we mainly need 2 analog outputs to control the speed of the motors. Our CPU S7-1512C-1 PN does have 2 analog output so we don't need to add a module.

GRAPH PLC_1	0	1			CPU 1512C-1 PN	6ES7 512-1CK01-0AB0	V2.9
AI 5/AQ 2_1	0	1 8	0...9	0...3	AI 5/AQ 2		
DI 16/DQ 16_1	0	1 9	10...11	4...5	DI 16/DQ 16		
DI 16/DQ 16_2	0	1 10	12...13	6...7	DI 16/DQ 16		
HSC_1	0	1 16	14...29	8...19	HSC		
HSC_2	0	1 17	30...45	20...31	HSC		
HSC_3	0	1 18	46...61	32...43	HSC		
HSC_4	0	1 19	62...77	44...55	HSC		
HSC_5	0	1 20	78...93	56...67	HSC		
HSC_6	0	1 21	94...109	68...79	HSC		
Pulse_1	0	1 32	110...113	80...91	PWM		
Pulse_2	0	1 33	114...117	92...103	PWM		
Pulse_3	0	1 34	118...121	104...115	PWM		
Pulse_4	0	1 35	122...125	116...127	PWM		
PROFINET interface_1	0	1 X1			PROFINET interface		
DI 16x24VDC BA_1	0	2	128...129		DI 16x24VDC BA	6ES7 521-1BH10-0AA0	V1.1
DQ 8x24VDC/2A HF_1	0	3		128	DQ 8x24VDC/2A HF	6ES7 522-1BF00-0AB0	V2.2

Figure 73: List of Main Components



Figure 74: Modules Principales

3.2.4 Choice of HMI lectern

In order to allow the monitor to select the choice of HMI, I opted for a cost-effective solution

3.2.4.1 WinCC Definition

WinCC is a software for Supervisory Control And Data Acquisition(SCADA), a term that describes the basic functions of a SCADA system. Companies use WinCC systems to control equipment across all their sites, but also to collect and record data about their operations.

3.2.4.2 How a WinCC system works

A WinCC is typically delivered as software combined with hardware elements, such as programmable logic controllers (PLCs) and remote terminal units (RTUs).
With WinCC systems, organizations can control their industrial processes either on-site or remotely, and interact directly with equipment, such as motors, pumps, and sensors, at all levels from a central location. These systems can sometimes automatically control equipment based on the data received.

WinCC systems also allow organizations to monitor their processes, report on them based on real-time data, and archive that data for further processing and evaluation. [4]

3.2.4.3 Applications WinCC

Organizations can use WinCC systems to:

- Control processes on-site or remotely
- Interact with devices using HMI software
- Collecting, monitoring and processing data
- Log events and data.

3.2.4.4 Core WinCC Network

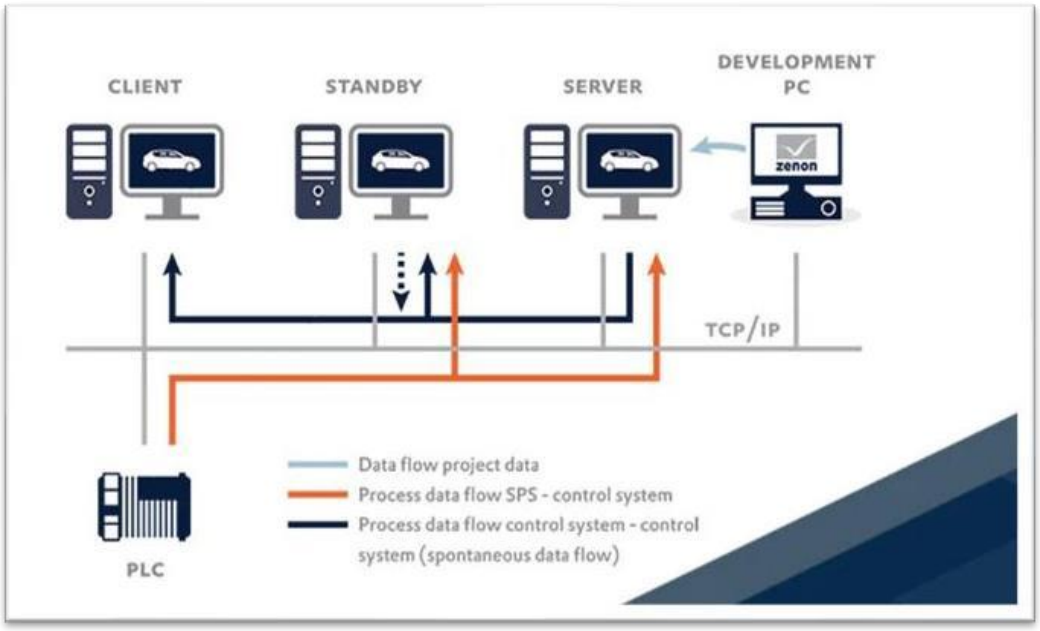


Figure 75: Basic WinCC binding

4. Programming on TIA portal

4.1 TIA Portal Definition

The **"Totally Integrated Automation Portal"** (**TIA**) software is a software for programming the PLCs of the **"S7-1500" series** and the operator panels of the KTP series.

This consists of STEP 7 Basic and WinCC Basic. It takes the same programming philosophy as the STEP 7 Pro software with a simplified interface and the integration of operator console programming. SIEMENS' goal is to integrate all of their product lines into a single software. When the TIA Portal is launched, the environment is broken down into two types of views: The Portal View and the Project View.



Figure 76: TIA Portal V17

4.1.1 Portal View

The portal view provides an overview of all project setup steps and guided access to your automation task.

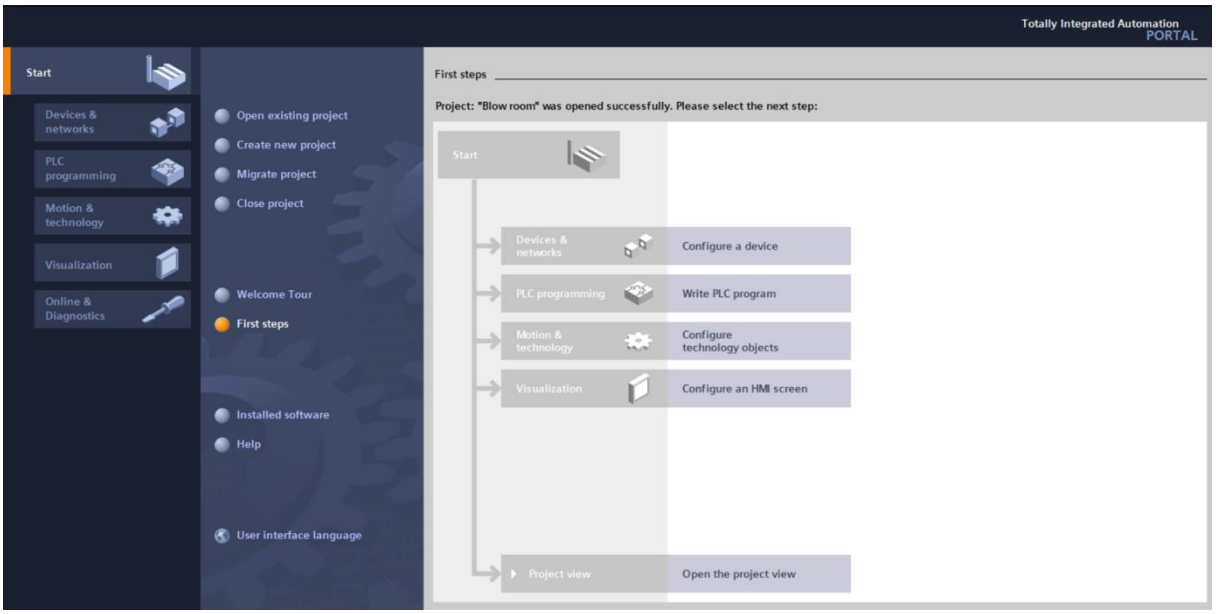


Figure 77: Portal View

4.1.2 Project View

The project view is a hierarchical structure view of a project's set of components that can be used to create all project goals and changes.

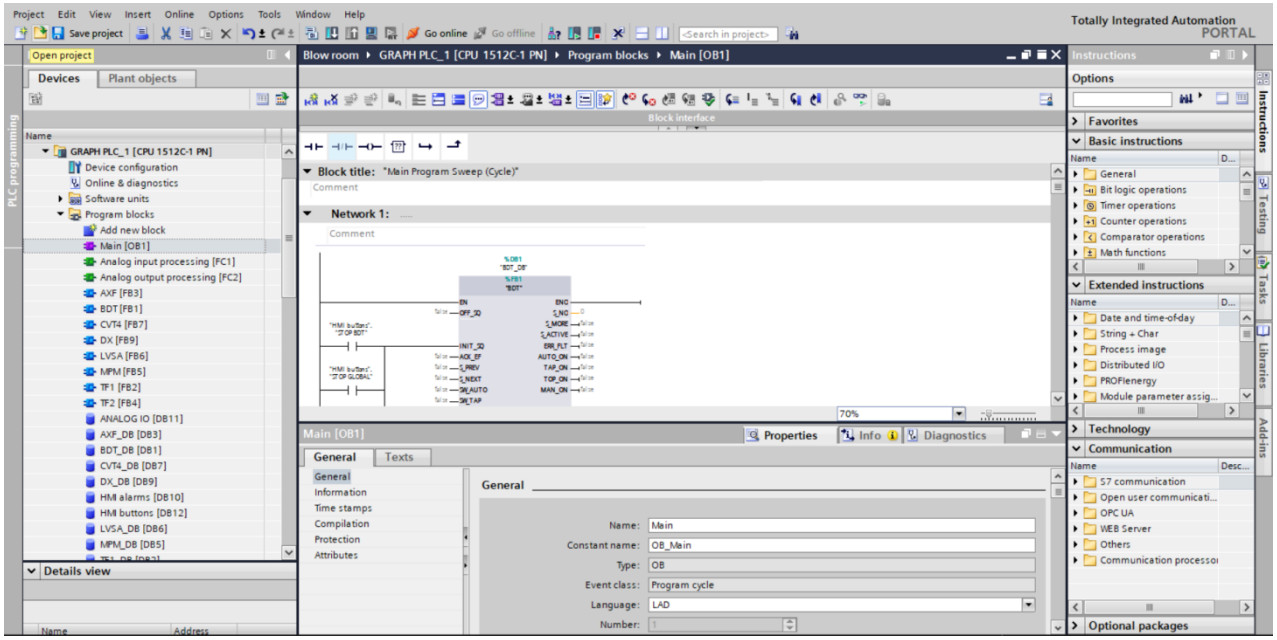


Figure 78: Project View

4.2 Creating Blocks

We have chosen to follow, for the programming of our application, the complex structured type which consists of the subdivision of the program into small parts, corresponding to the automation functions (**FC)** and (**FB**) that can be used by calling on them in the organizational block (**OB).**

We used these blocks for the programming of our machine:

4.2.1 Organization Block (OB1)

It is the interface between the user program and the operating system, i.e. the CPU executes only the instruction that is on that block. OB1 contains

the block call statement for the other FB, FC, and DB blocks to be executed by the PLC.

4.2.2 Function (FC)

A function is a block of code with no afterglow. After the functions are processed, the output parameters contain the calculated function values. It is then up to us to organize the use and backup of the effective parameters, according to our needs.

4.2.3 Function Block (FB)

The FB block needs a memory area assigned to it for each call (instance). It can be called by the OB block to structure the program into tasks and sub tasks. These variables are local variables.

4.2.4 Data Block (DB)

Global data frames store data that can be used by all other blocks. The maximum size of the DBs varies depending on the CPU

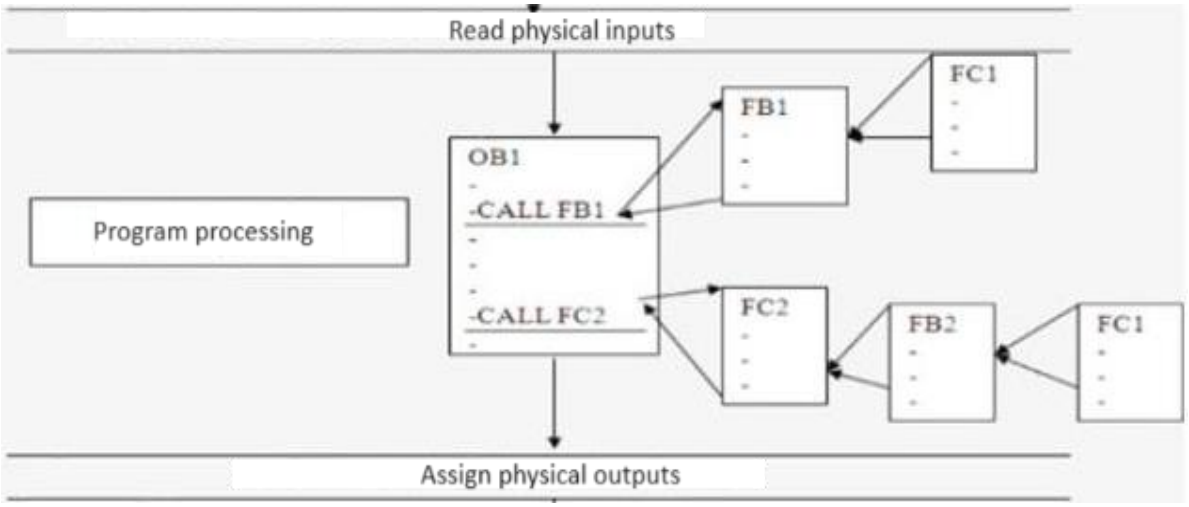


Figure 79: General structure of a programme

- To create a block you have to click on program block, then click on add a new block after choosing one of the blocks, the following window will appear:

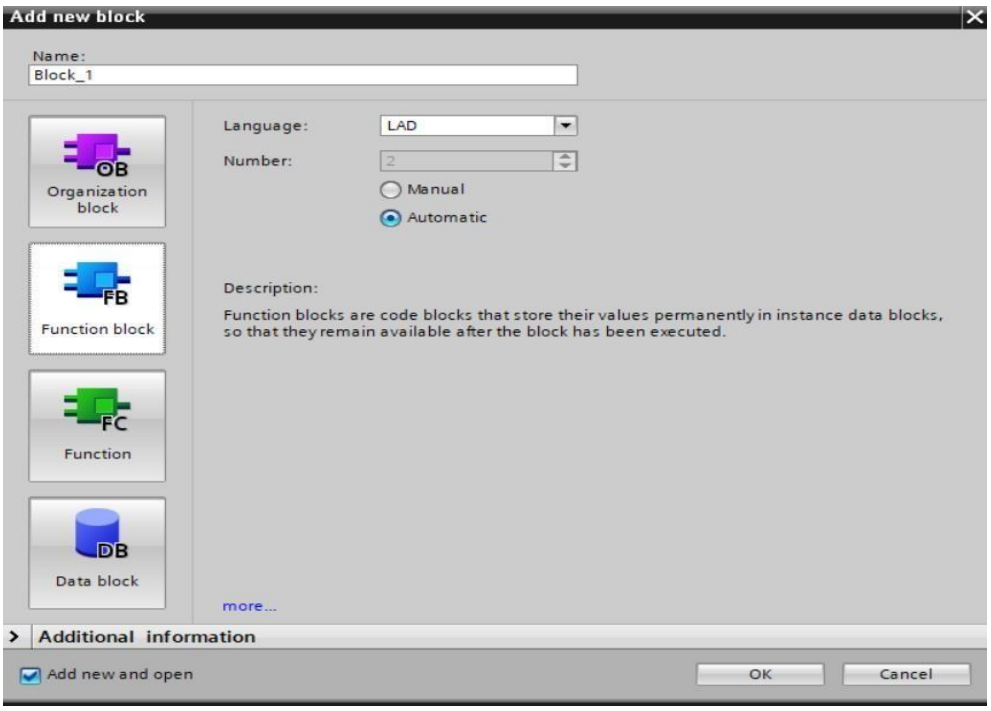


Figure 80: Block Selection Window

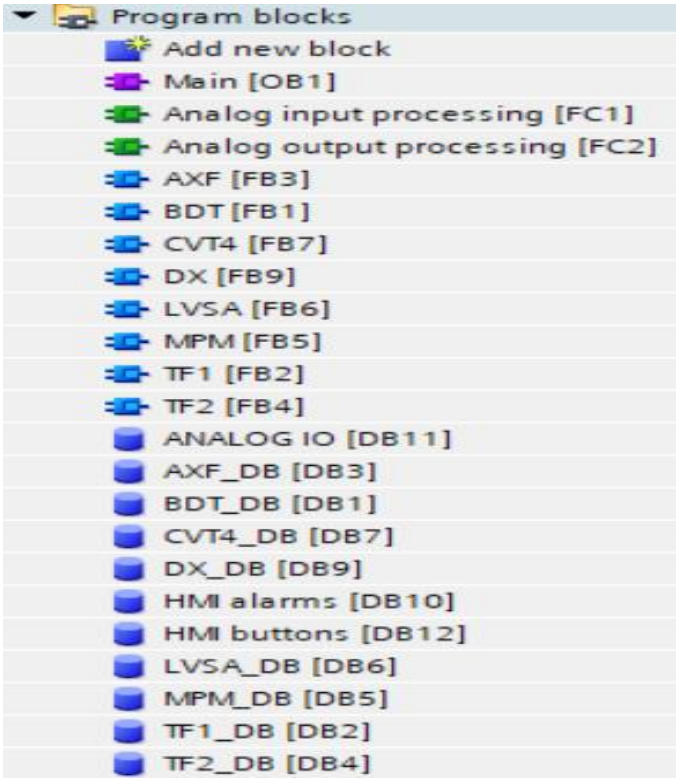


Figure 81: Block List of my program

4.3 Choice of programming language

The GRAPH language is very easy to use and ideal for visualizing and diagnosing programs during maintenance operations, so we use this language.

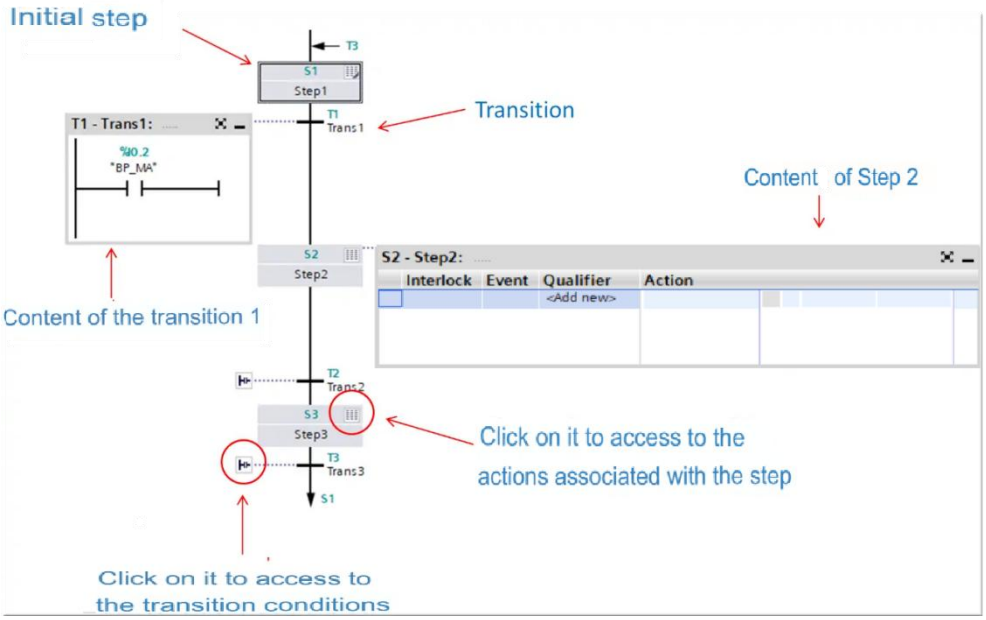


Figure 82: GRAPH Language

4.4 Programming System

See appendix

4.5 Example

Below is an easy example of how to transform the Technical fan 1 Grafcet to GRAPH language.

4.5.1 TF1 Block (FB2)

When the program starts, the Technical Fan 1 Grafcet is initialized using start button.

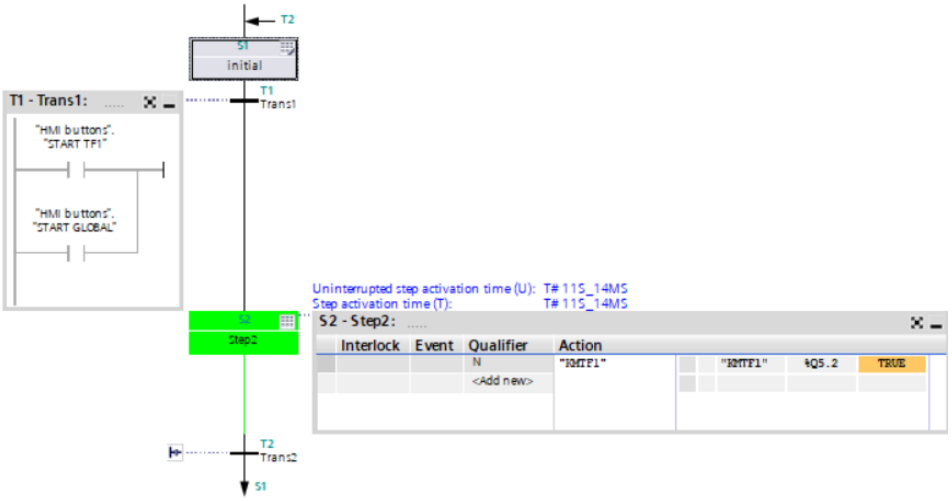


Figure 83: Initializing TF1 Grafcet

This Block has been called up in the **Main block** to be executed, the **INIT_SQ** is for initializing the sequence in case one of these inputs is triggered, its is usually used for **Security Grafcet**.

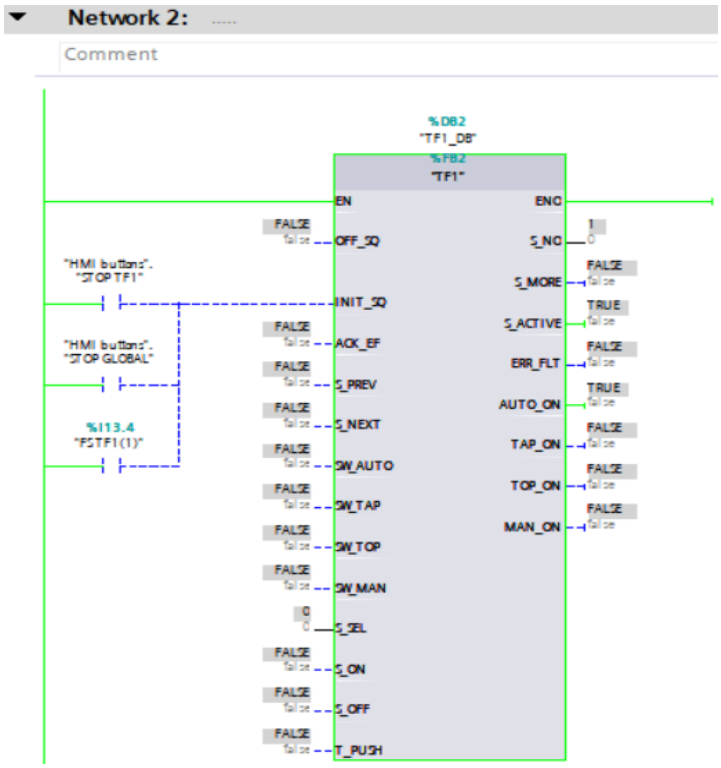


Figure 84: FB2 Call TF1 block in Main(OB1)

4.5.2 Analog INPUT processing (FC1)

This Function is used to convert analog input values (physical signals such as pressure) into digital values that the PLC can understand, So I have used 2 conversion operations for the analog input conversion, the first operation is **NORM_X** which is used to normalize the input value to a linear scale within the specified range of **[0 … 27648]** which represent the maximum **Integer** value of **SIEMENS**' CPUs, the second operation is **SCALE_X** which is used to take the normalized value and scale it proportionally to the specified range of **[0 … 2000Pa].**

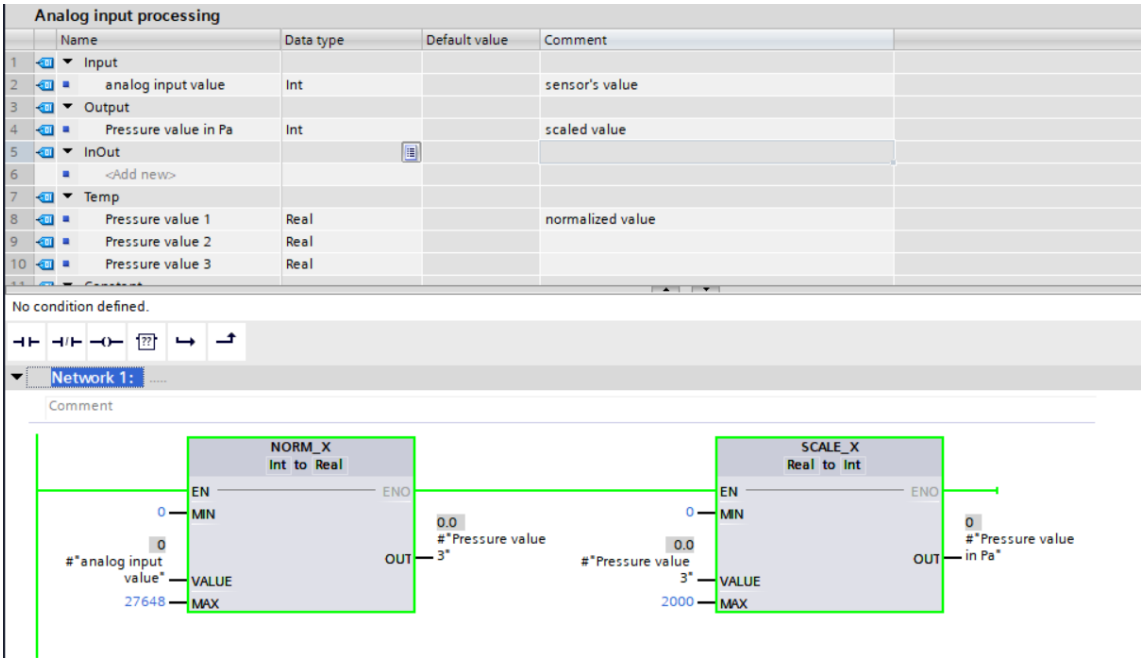


Figure 85: FC Block of analog Input processing

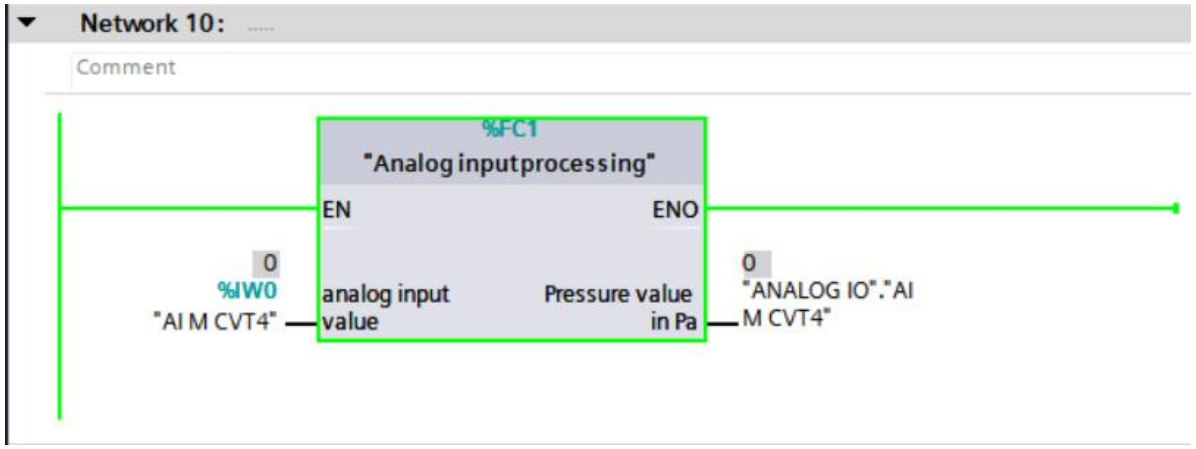


Figure 86: FC Call analog input in Main(OB1)

4.5.3 Analog OUTPUT processing (FC2)

This Function is used to convert the digital values into analog values to regulate the desired output value , So I have used 2 conversion operations for the analog input conversion, the first operation is **NORM_X** which is used to normalize the input value to a linear scale within the specified range of **[0 … 100%]** which represent the desired percentage I want the output to operate with , the second operation is **SCALE_X** which is used to take the normalized value and scale it proportionally to the specified range of **[0 … 32767]** which represent the maximum **Real** value of **SIEMENS**' CPUs.

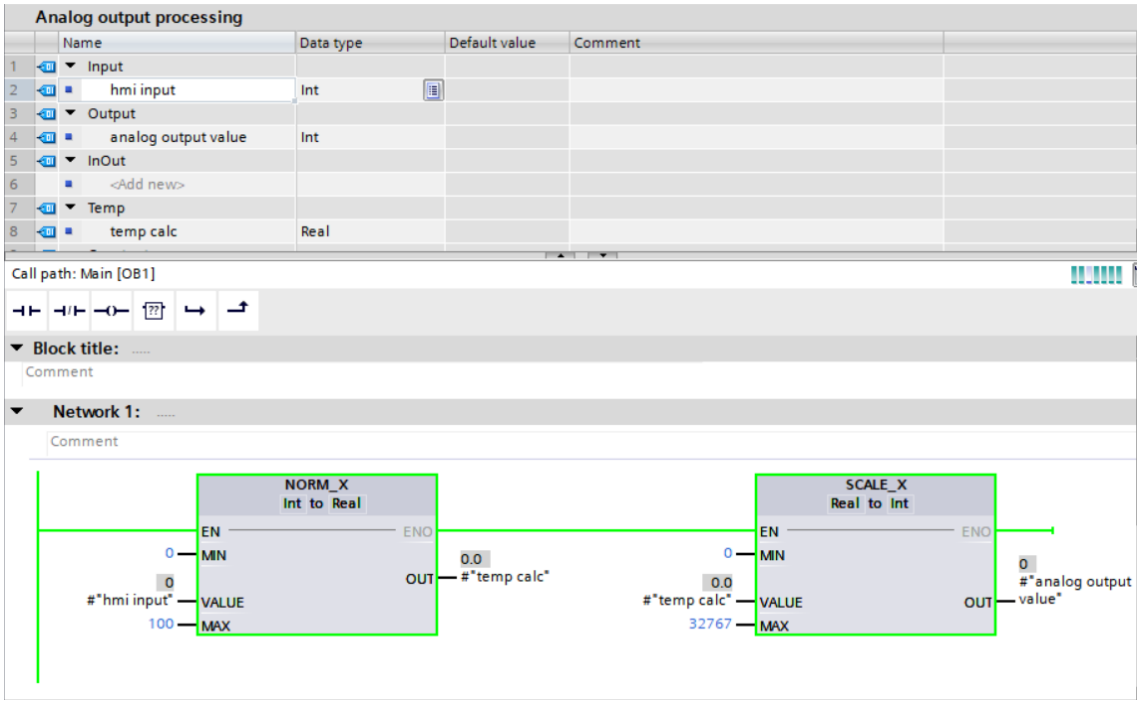


Figure 87: FC Block of analog Output processing

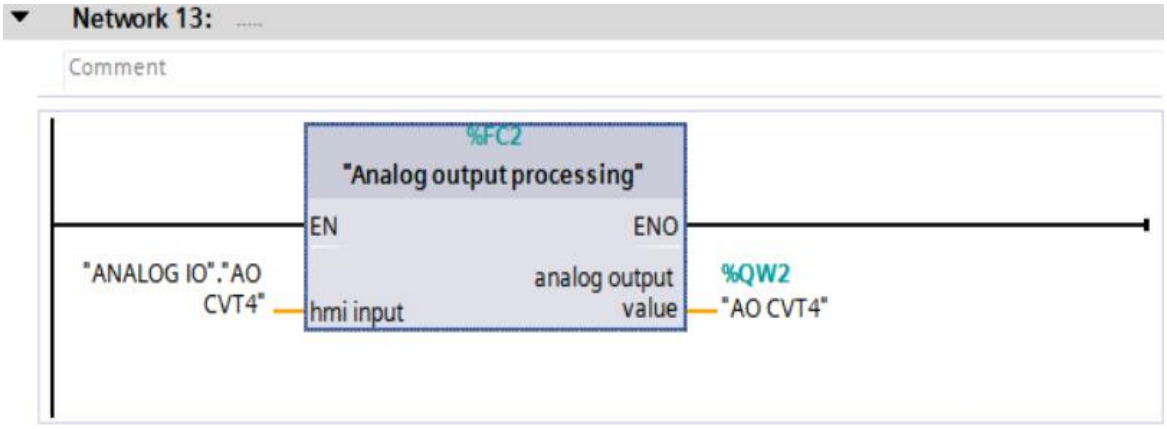


Figure 88: FC Call analog Output in Main(OB1)

4.6 Program Simulation

Once the programs have been completed, **TIAPORTAL** allows the simulation of the configuration and the program implemented in the **"CPU 1512C-1 PN"** of the machine thanks to its extension

"S7-PLCSIM". The "S7-PLCSIM" **simulator application** has an interface allowing the visualization and forcing of the different states of the inputs and outputs of the PLC.



Figure 89: Simulation bar Tia Portal

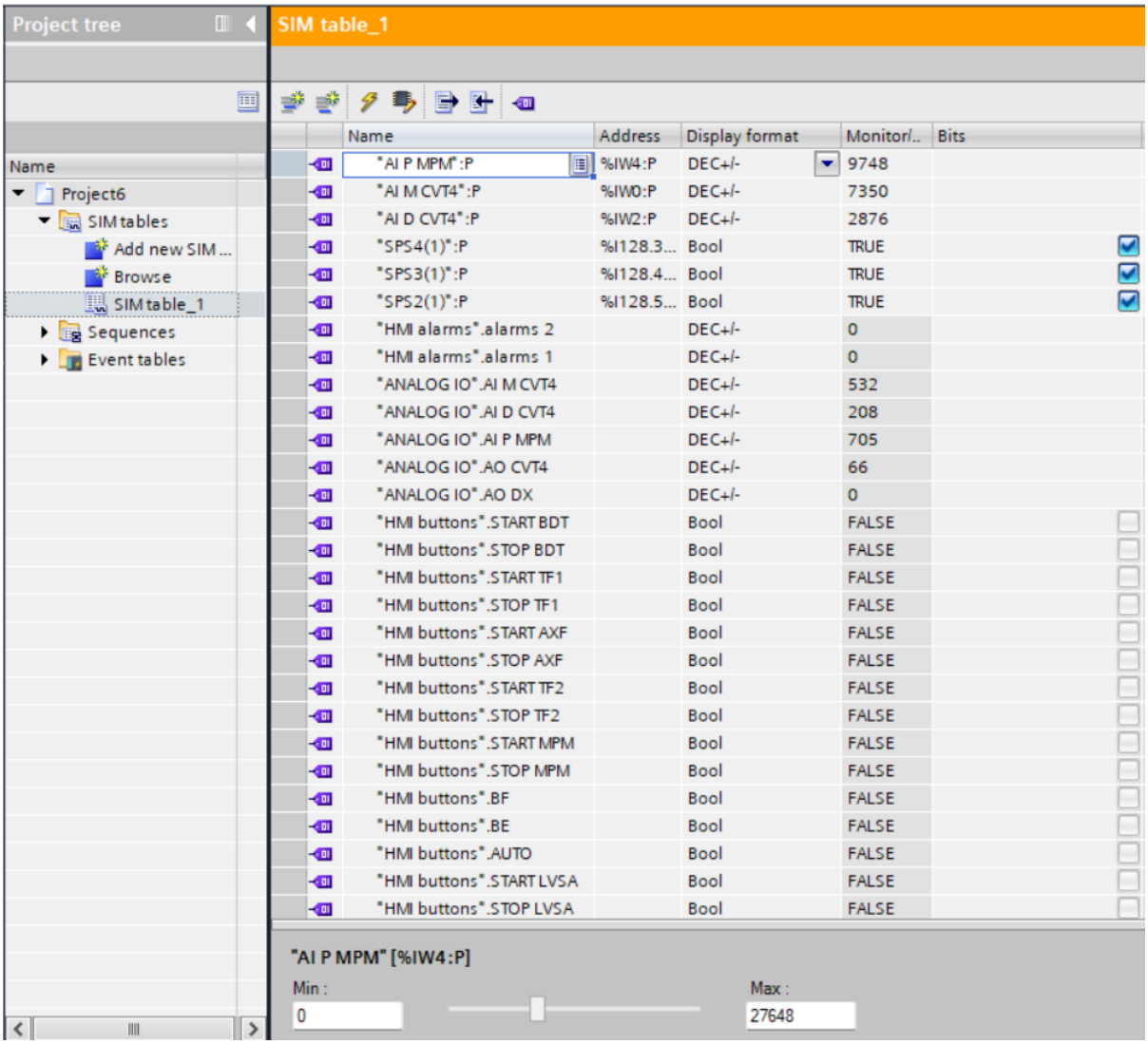


Figure 90: PLC SIM TABLE

4.7 HMI Connection & PLC

To use a WinCC HMI, you need to establish a connection between your CPU and a HMI system.

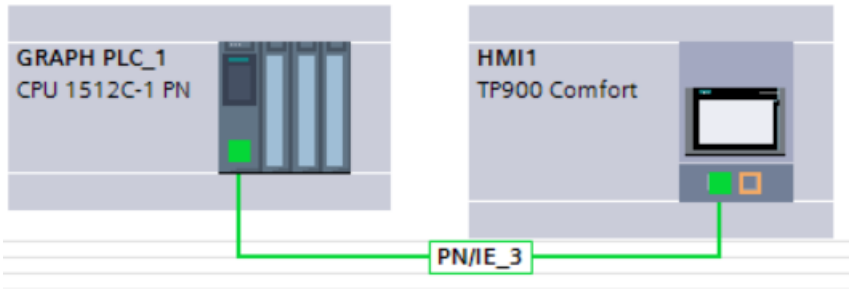


Figure 91: PLC-HMI link

4.8 HMI Tags

The HMI Tags are associated with the PLC Tags to ensure that the inputs and outputs are assigned to the same memory location .

HMI tags

Name	Tag table	Data type	Connection	PLC name	PLC tag	Address	Access mode	Acquisition cycle
ANALOG IO_AI D CVT4	Default tag table	Int	HMI_Connection_3	GRAPH PLC_1	"ANALOG IO"."AI D CVT4"		<symbolic access>	100 ms
ANALOG IO_AI M CVT4	Default tag table	Int	HMI_Connection_3	GRAPH PLC_1	"ANALOG IO"."AI M CVT4"		<symbolic access>	100 ms
ANALOG IO_AI P MPM	Default tag table	Int	HMI_Connection_3	GRAPH PLC_1	"ANALOG IO"."AI P MPM"		<symbolic access>	100 ms
ANALOG IO_AO CVT4	Default tag table	Int	HMI_Connection_3	GRAPH PLC_1	"ANALOG IO"."AO CVT4"		<symbolic access>	100 ms
ANALOG IO_AO DX	Default tag table	Int	HMI_Connection_3	GRAPH PLC_1	"ANALOG IO"."AO DX"		<symbolic access>	100 ms
HMI alarms_alarms 1	Default tag table	Int	HMI_Connection_3	GRAPH PLC_1	"HMI alarms"."alarms 1"		<symbolic access>	100 ms
HMI alarms_alarms 2	Default tag table	Int	HMI_Connection_3	GRAPH PLC_1	"HMI alarms"."alarms 2"		<symbolic access>	100 ms
HMI communication_AUTO	Default tag table	Bool	HMI_Connection_3	GRAPH PLC_1	"HMI buttons".AUTO		<symbolic access>	100 ms
HMI communication_BE	Default tag table	Bool	HMI_Connection_3	GRAPH PLC_1	"HMI buttons".BE		<symbolic access>	100 ms
HMI communication_BF	Default tag table	Bool	HMI_Connection_3	GRAPH PLC_1	"HMI buttons".BF		<symbolic access>	100 ms
HMI communication_START AXF	Default tag table	Bool	HMI_Connection_3	GRAPH PLC_1	"HMI buttons"."START AXF"		<symbolic access>	100 ms
HMI communication_START BDT	Default tag table	Bool	HMI_Connection_3	GRAPH PLC_1	"HMI buttons"."START BDT"		<symbolic access>	100 ms
HMI communication_START CVT4	Default tag table	Bool	HMI_Connection_3	GRAPH PLC_1	"HMI buttons"."START CV...		<symbolic access>	100 ms
HMI communication_START DX	Default tag table	Bool	HMI_Connection_3	GRAPH PLC_1	"HMI buttons"."START DX"		<symbolic access>	100 ms
HMI communication_START GLOBAL	Default tag table	Bool	HMI_Connection_3	GRAPH PLC_1	"HMI buttons"."START GL...		<symbolic access>	100 ms
HMI communication_START LVSA	Default tag table	Bool	HMI_Connection_3	GRAPH PLC_1	"HMI buttons"."START LV...		<symbolic access>	100 ms
HMI communication_START MPM	Default tag table	Bool	HMI_Connection_3	GRAPH PLC_1	"HMI buttons"."START MP...		<symbolic access>	100 ms
HMI communication_START TF1	Default tag table	Bool	HMI_Connection_3	GRAPH PLC_1	"HMI buttons"."START TF1"		<symbolic access>	100 ms
HMI communication_START TF2	Default tag table	Bool	HMI_Connection_3	GRAPH PLC_1	"HMI buttons"."START TF2"		<symbolic access>	100 ms
HMI communication_STOP AXF	Default tag table	Bool	HMI_Connection_3	GRAPH PLC_1	"HMI buttons"."STOP AXF"		<symbolic access>	100 ms
HMI communication_STOP BDT	Default tag table	Bool	HMI_Connection_3	GRAPH PLC_1	"HMI buttons"."STOP BDT"		<symbolic access>	100 ms
HMI communication_STOP CVT4	Default tag table	Bool	HMI_Connection_3	GRAPH PLC_1	"HMI buttons"."STOP CVT4"		<symbolic access>	100 ms
HMI communication_STOP DX	Default tag table	Bool	HMI_Connection_3	GRAPH PLC_1	"HMI buttons"."STOP DX"		<symbolic access>	100 ms
HMI communication_STOP GLOBAL	Default tag table	Bool	HMI_Connection_3	GRAPH PLC_1	"HMI buttons"."STOP GLO...		<symbolic access>	100 ms
HMI communication_STOP LVSA	Default tag table	Bool	HMI_Connection_3	GRAPH PLC_1	"HMI buttons"."STOP LVSA"		<symbolic access>	100 ms
HMI communication_STOP MPM	Default tag table	Bool	HMI_Connection_3	GRAPH PLC_1	"HMI buttons"."STOP MPM"		<symbolic access>	100 ms
HMI communication_STOP TF1	Default tag table	Bool	HMI_Connection_3	GRAPH PLC_1	"HMI buttons"."STOP TF1"		<symbolic access>	100 ms
HMI communication_STOP TF2	Default tag table	Bool	HMI_Connection_3	GRAPH PLC_1	"HMI buttons"."STOP TF2"		<symbolic access>	100 ms

Figure 92: HMI Tags

4.9 HMI View

In this section, we will present the HMI interface that I have made for the supervision and control of the operation of the Blow room line machines.

4.9.1 Root screen View

The view of the Root screen is represented by the figure Below:

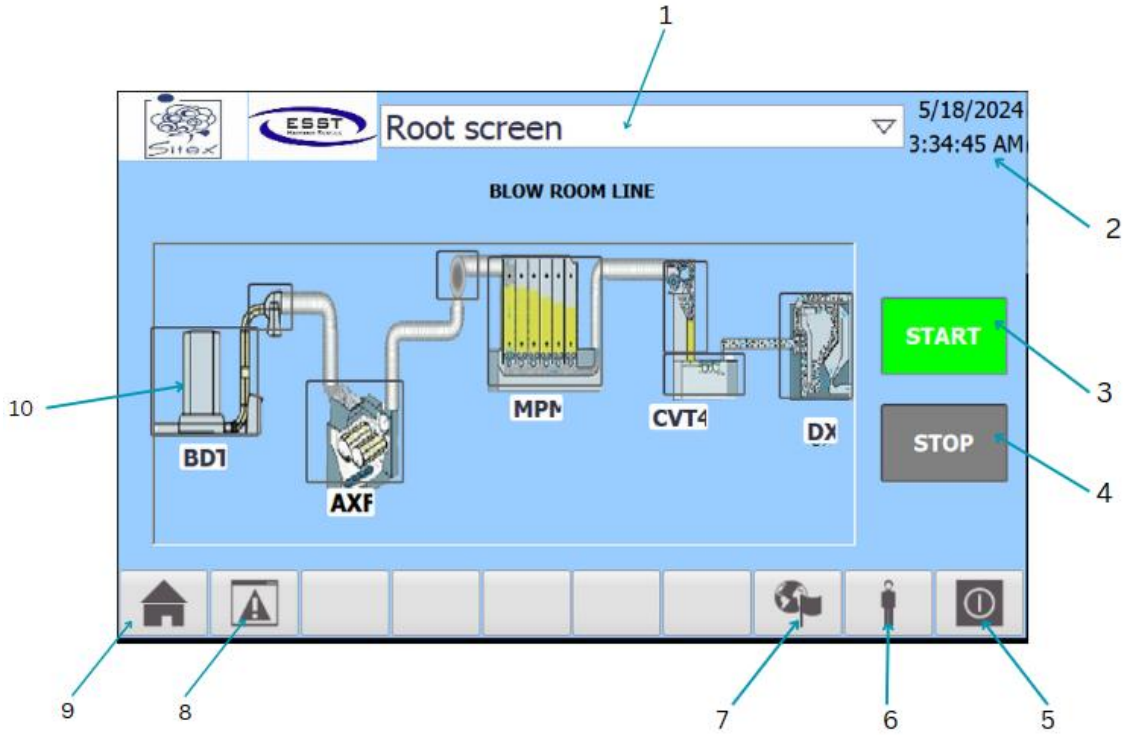


Figure 93: Home view

Table 11: Main Elements

N°	Name
1	Navigation bar
2	Date & Time
3	Global Start Button
4	Global Stop Button
5	Shutdown Control
6	Administrator Access
7	Language preference
8	Signal Alarm
9	Home Page
10	Select machine

4.9.2 BDT View

The view below shows the operation of BDT, when the motors are started, the Arrows and LEDS turn on green indicating that the motors are running smoothly.

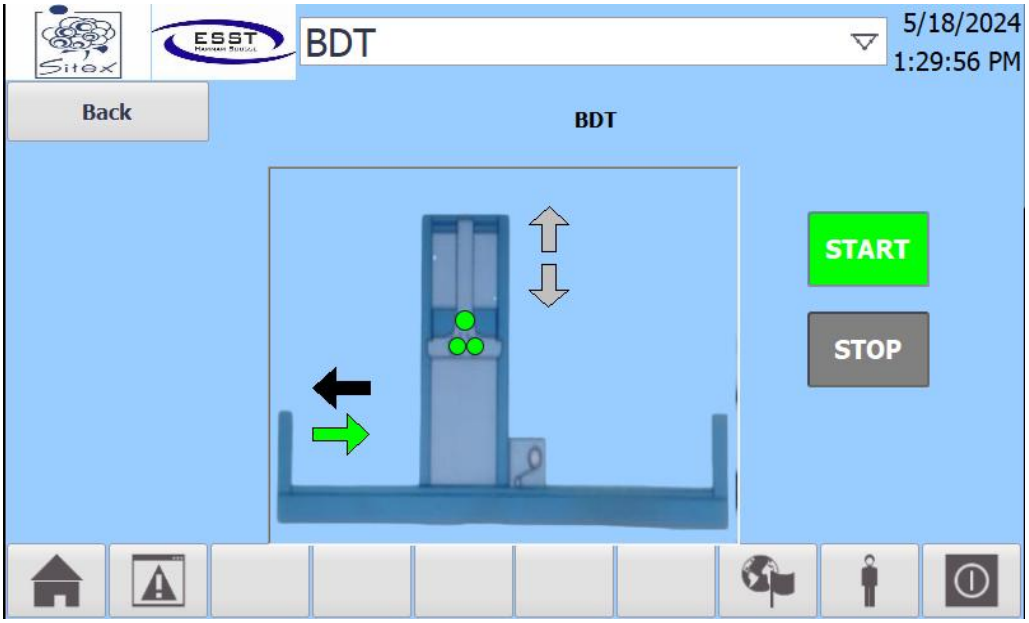


Figure 94: BDT View

4.9.3 TF1 View

The view below shows a simple view of Technical Fan 1.

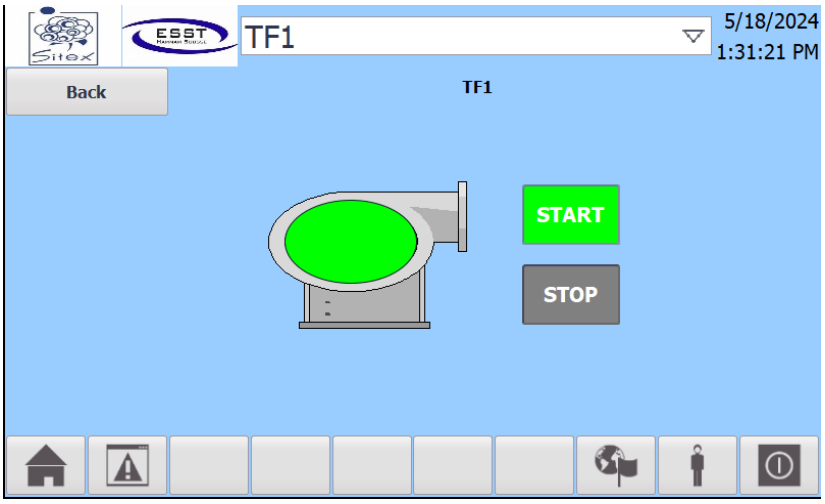


Figure 95: TF1 View

4.9.4 AXF View

The view below shows the operation of Axiflow and its motors running smoothly.

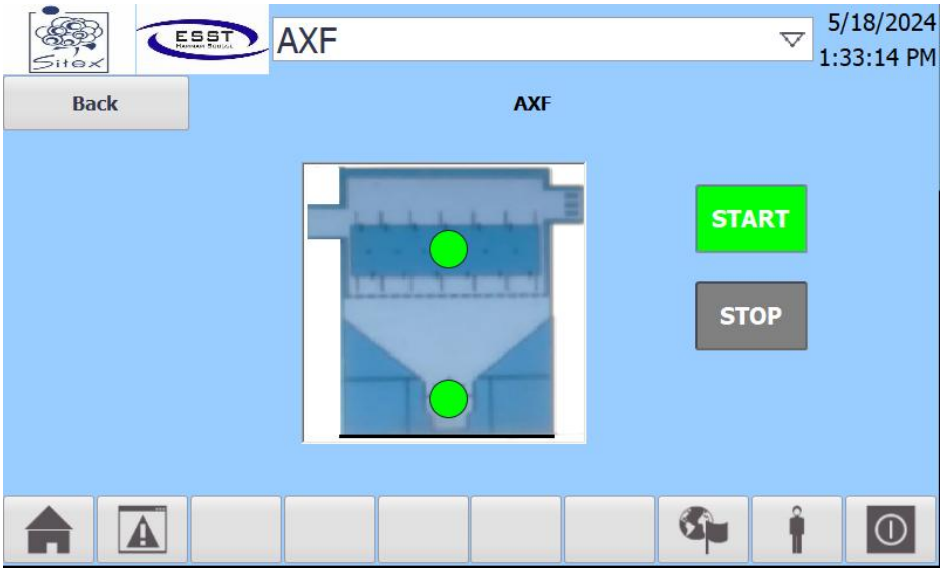


Figure 96: AXIFLOW View

4.9.5 TF2 View

The view below shows a simple view of Technical Fan 1.

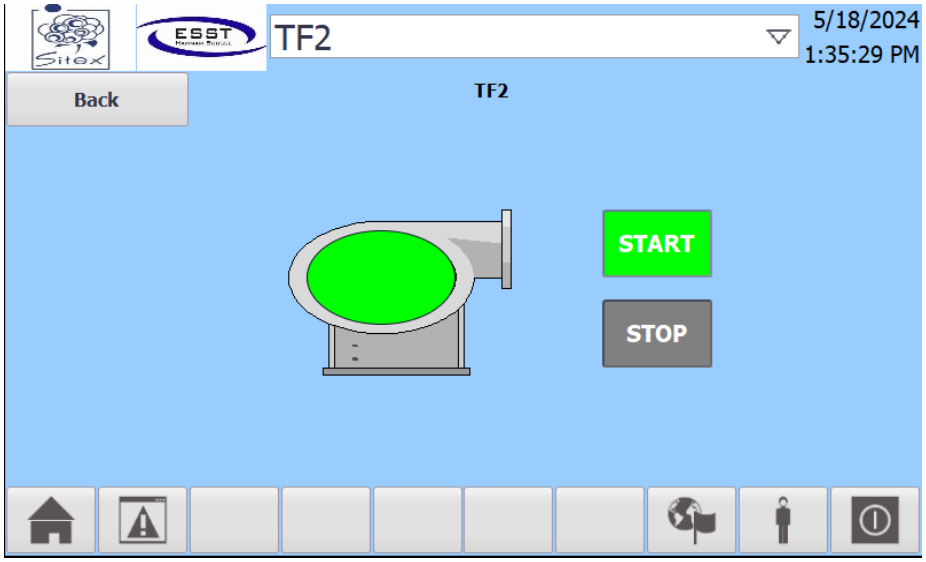


Figure 97: TF1 View

4.9.6 MPM View

The view below shows the mechanism of MPM and how to choose between different operations. For example pressing start button then choosing automatic mode , In addition to that there is a pressure sensor which calculates the pressure in the machine.

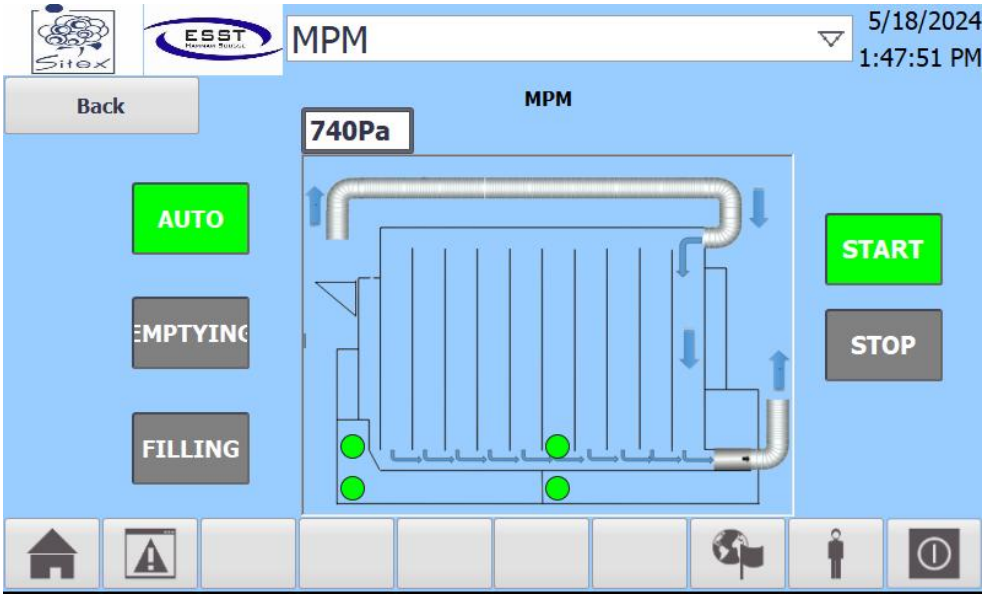


Figure 98: MPM View

4.9.7 LVSA View

The view below shows the operation of LVSA and its motors running smoothly.

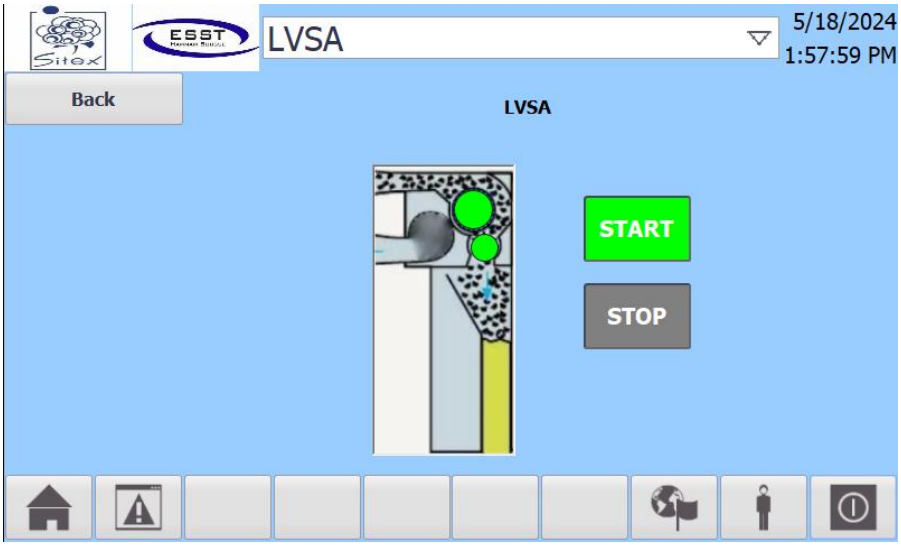


Figure 99: LVSA View

4.9.8 CVT4 View

The view below shows the mechanism of CVT4 and how to control the speed of the carpet motor on scale of 0 to 100%, In addition to that there is 2 pressure sensors which calculates the pressure of cotton and dust gathered in the machine.

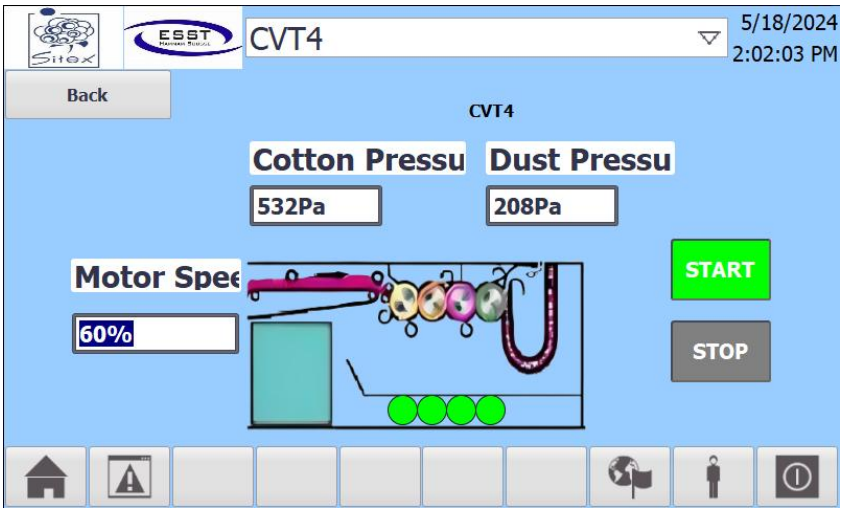


Figure 100: CVT4 View

4.9.9 DX View

The view below shows the mechanism of DX and how to control the speed of the first motor on scale of 0 to 100% , the 2nd and 3rd motors are running smoothly.

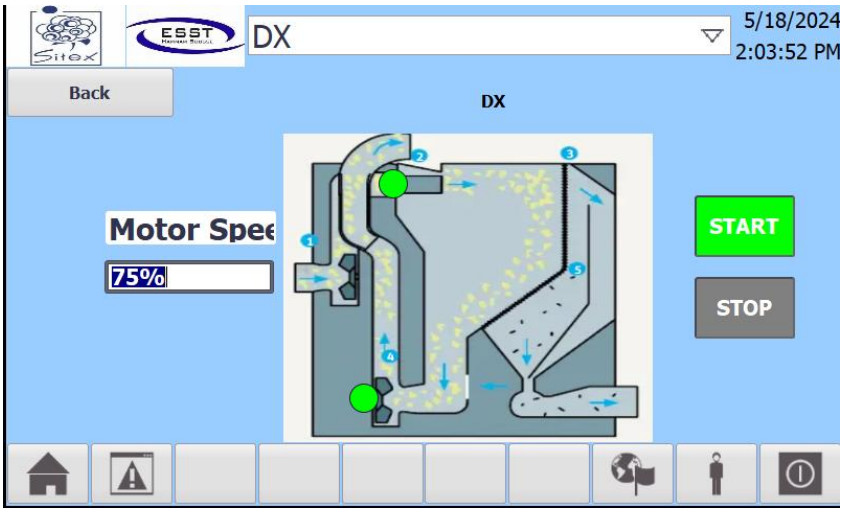


Figure 101: DX View

4.9.10 Alarm View

When an error or fault is detected, the user can find all of system errors in detail in this page.

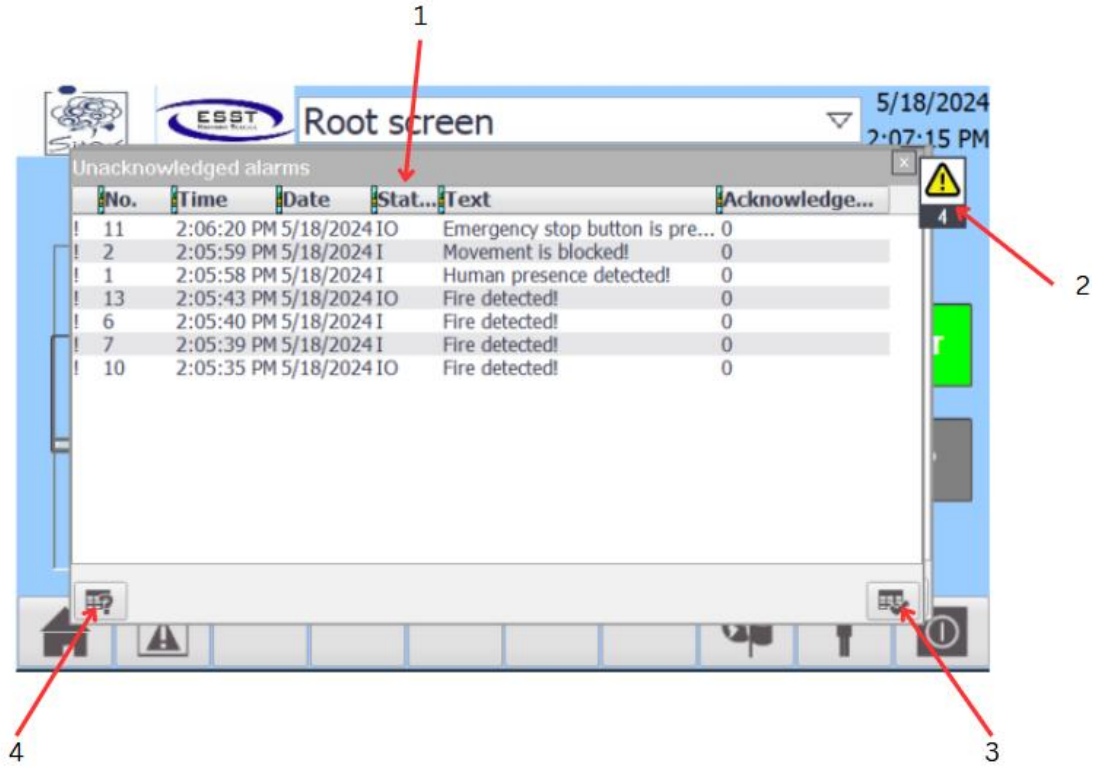


Figure 102: Alarm View

Table 10: Main Elements of the Root screen

N°	Name
1	Status of the error it is fixed or not
2	Number of unfixed alarms
3	The fixed alarms need to be acknowledge
4	Addition info text on where the error happened

5. Conclusion

In this chapter, we presented the new changes that have been added to our system, and then we developed the Grafcet of the system in different levels.

Thus, we also did the simulation of the program and the Human Machine Interface (**HMI**) using the software **"TIA Portal V17"** and its **tools "S7_PLCSIM"** and **"WINCC".**

Overall Conclusion and Outlook

My end-of-studies project is carried out within the company **SITEX**.

The objective of this internship is to automate and supervise Blow room line.

This project was also an opportunity to handle the various existing equipment in the company and to monitor the different tasks carried out in this direction.

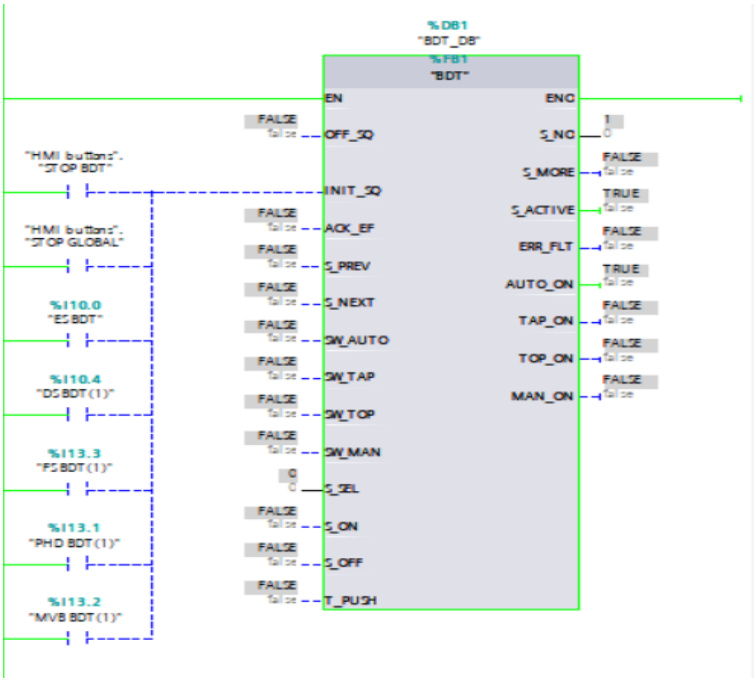
The goal of my work was to transform machines based on wired logic into machines that runs on a PLC **"S7-1500"** and program it on the TIA PORTAL programming software. Create an HMI user interface in WinCC . I have verified the proper functioning of our system, which is done by the **SIMATIC S7-PLCSIM simulator**.

Finally, it is necessary to state that this project can be considered as a real opportunity to improve , on the one hand my competence in the field of automation and on the other hand, familiarization with the techniques and tools of industrial computing. Indeed, this project allowed me to gain professional and social experience and to know how to integrate into a working group by exchanging ideas and points of view.

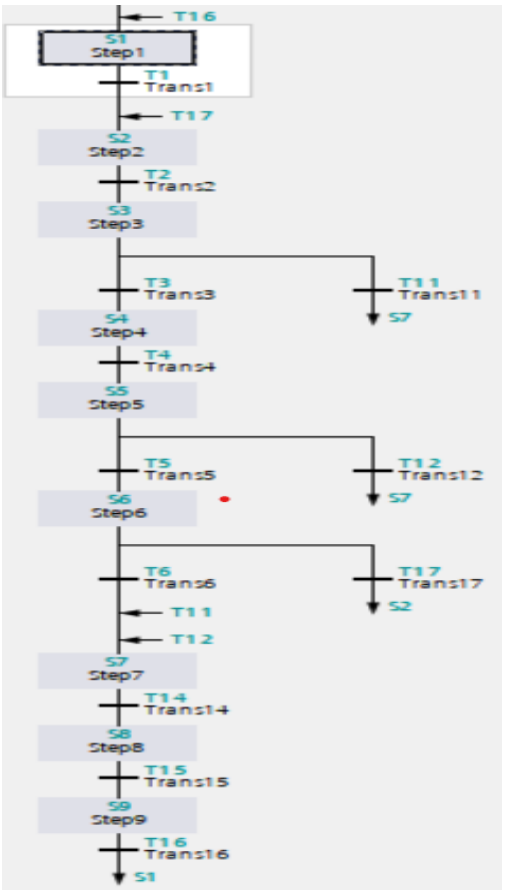
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BDT Block FB1 ..... Appendix 1
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MPM Block FB5 ..... Appendix 5
LVSA Block FB6 ..... Appendix 6
CVT4 Block FB7 ..... Appendix 7
DX Block FB9 ..... Appendix 8
Block DB10 ..... Appendix 9
Block FC1 ..... Appendix 10
Block FC2 ..... Appendix 11

Appendix 1: BDT Block FB1

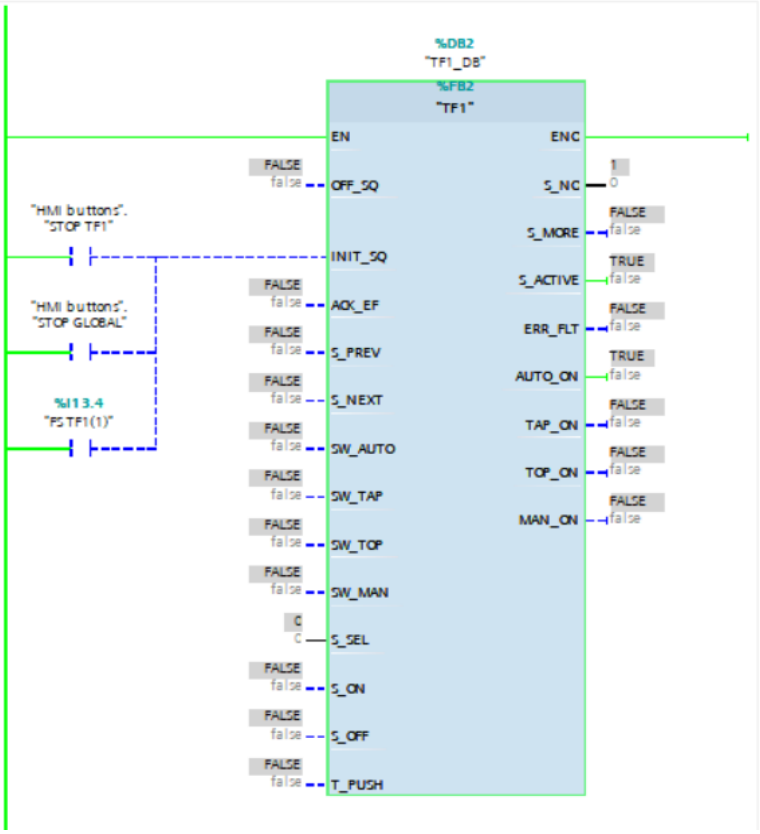


FB1 Call (BDT) in Main OB1

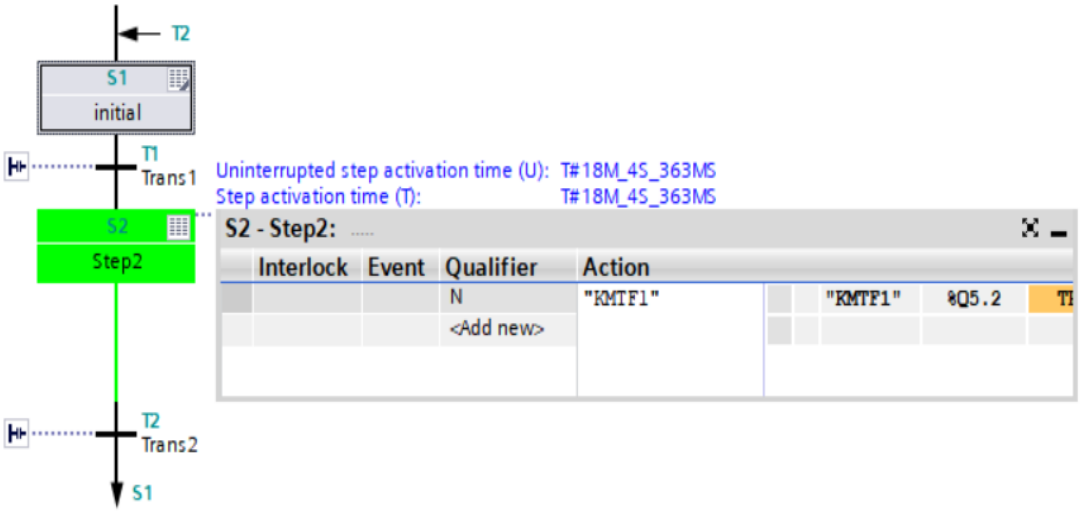


Grafcet BDT

Appendix 2: TF1 Block FB2

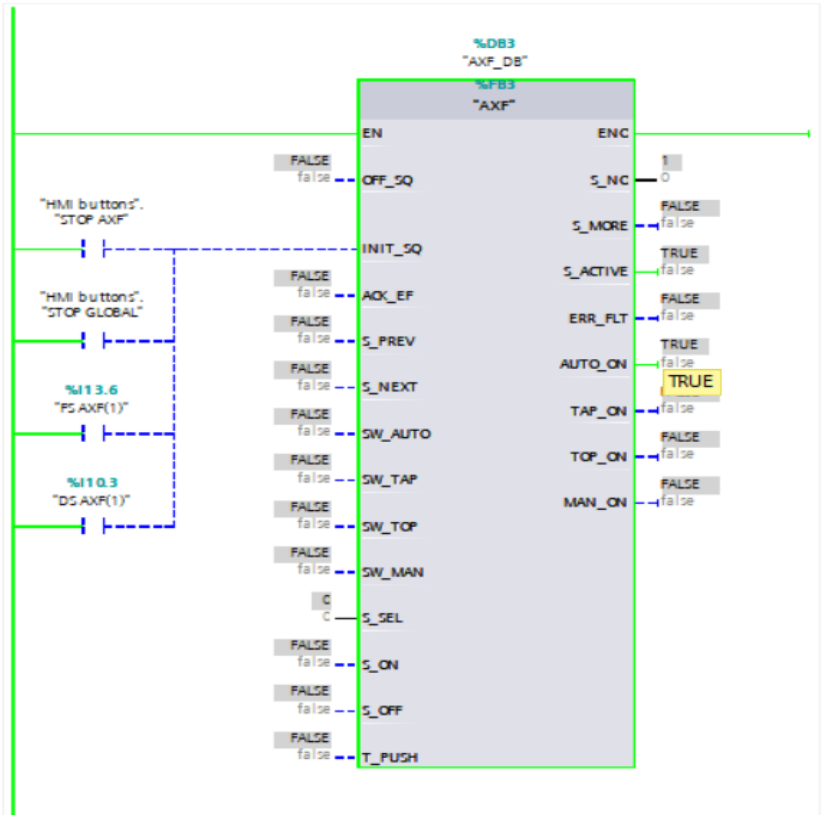


FB2 Call (TF1) in Main OB1



Grafcet TF1

Appendix 3: AXF Block FB3

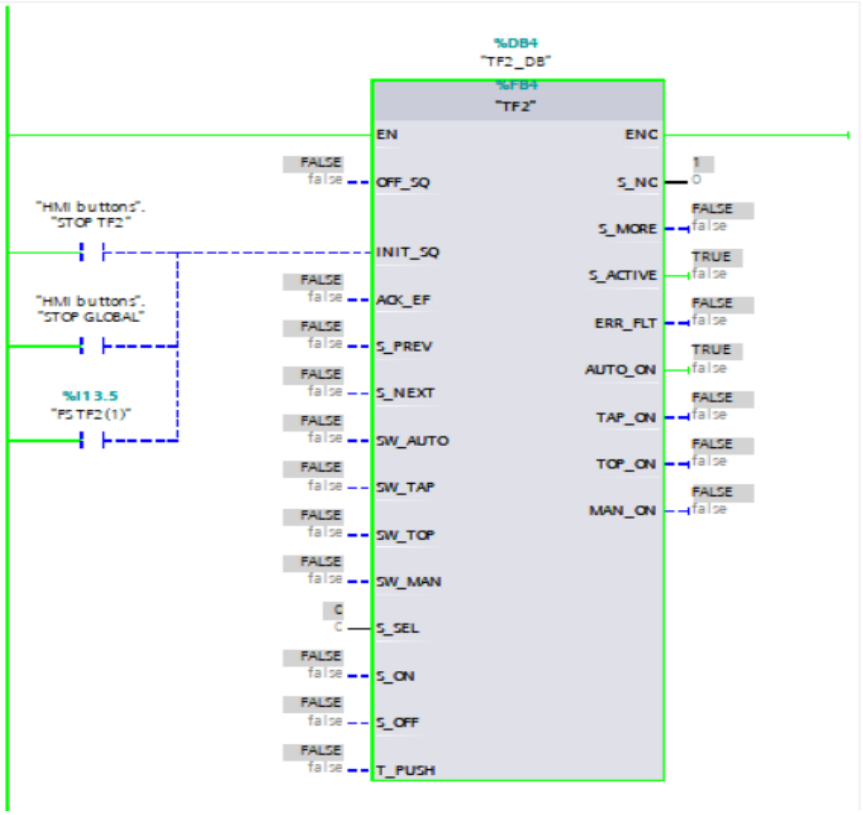


FB3 Call (AXF) in Main OB1

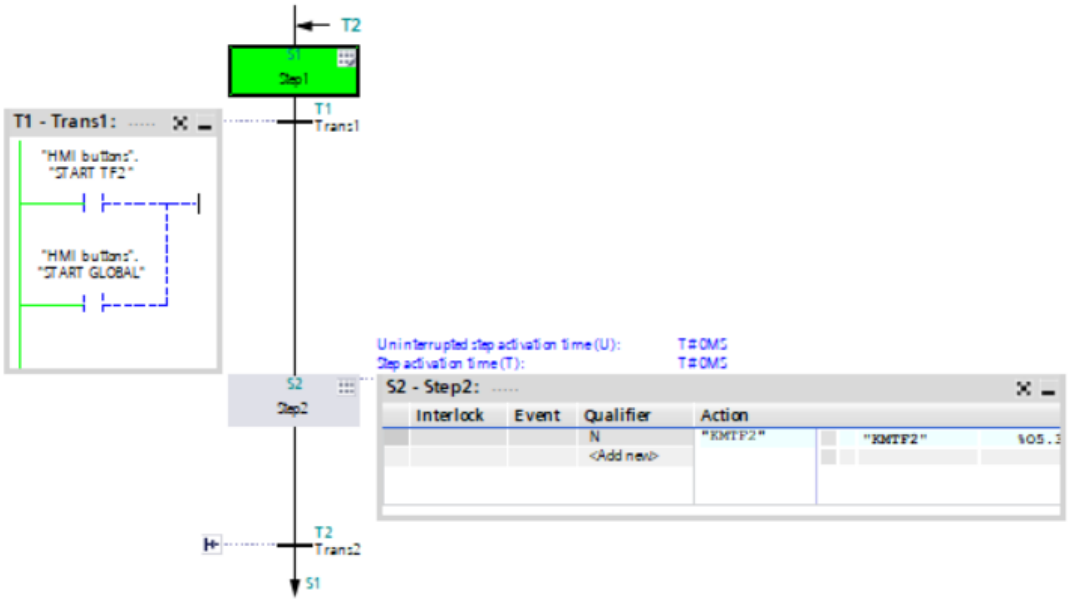


AXF Grafcet

Appendix 4: TF2 Block FB4

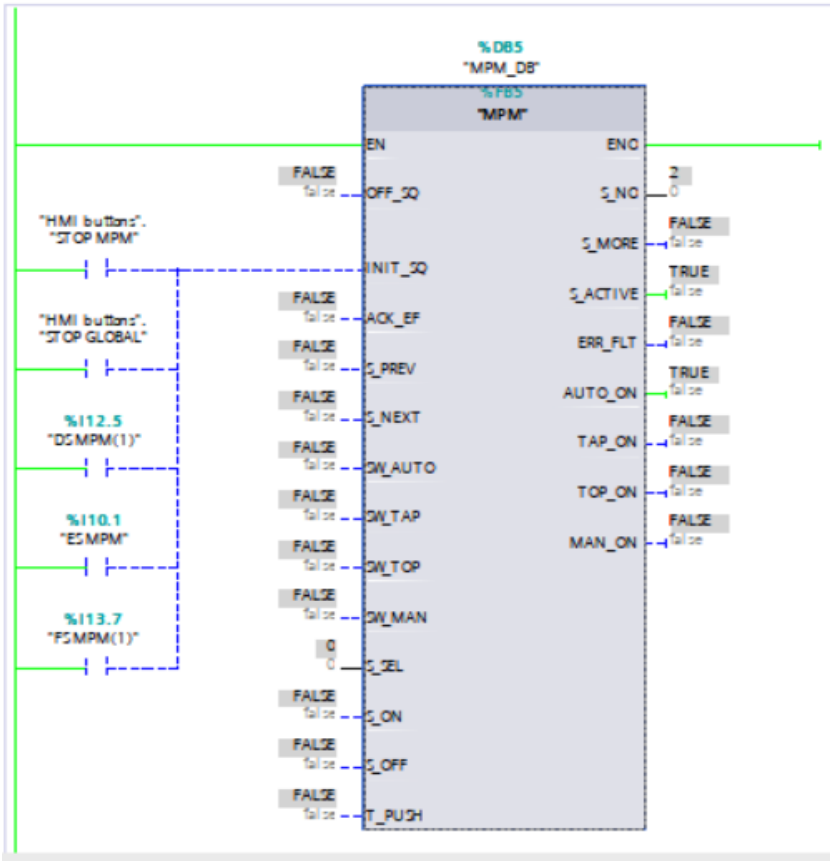


FB4 Call (TF2) in Main OB1

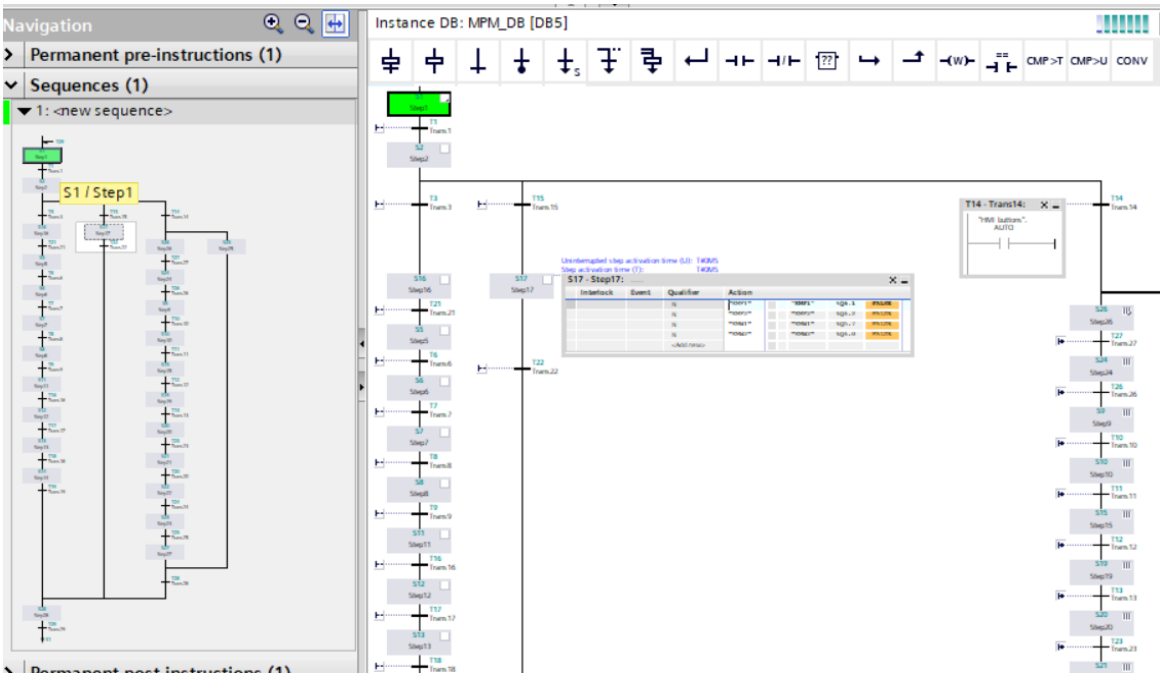


Grafcet TF2

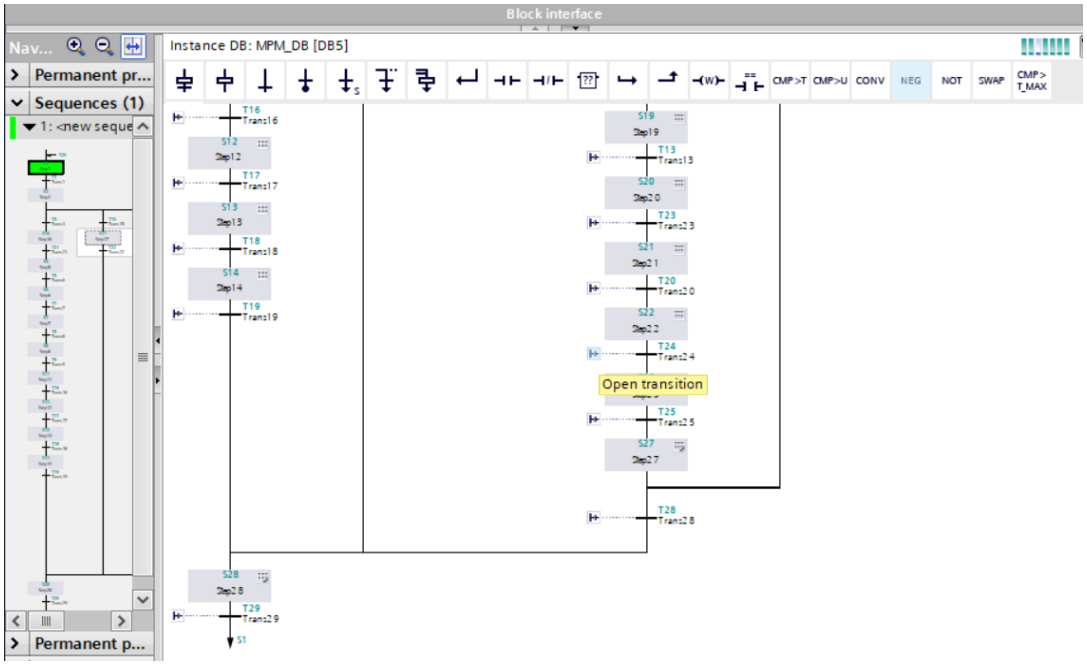
Appendix 5: MPM Block FB5



FB5 Call (MPM) in Main OB1

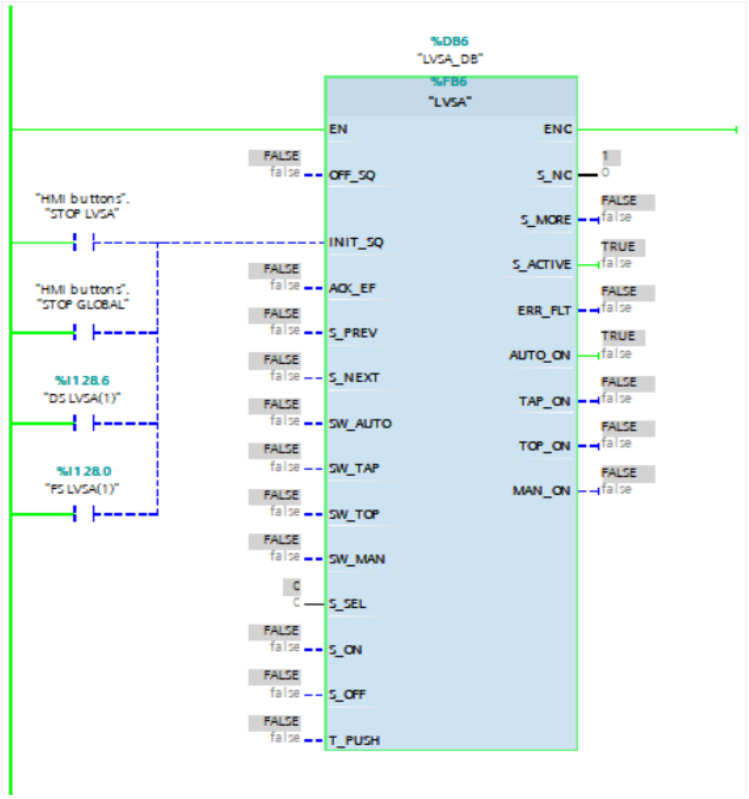


Automation and supervision of Blow Room Line

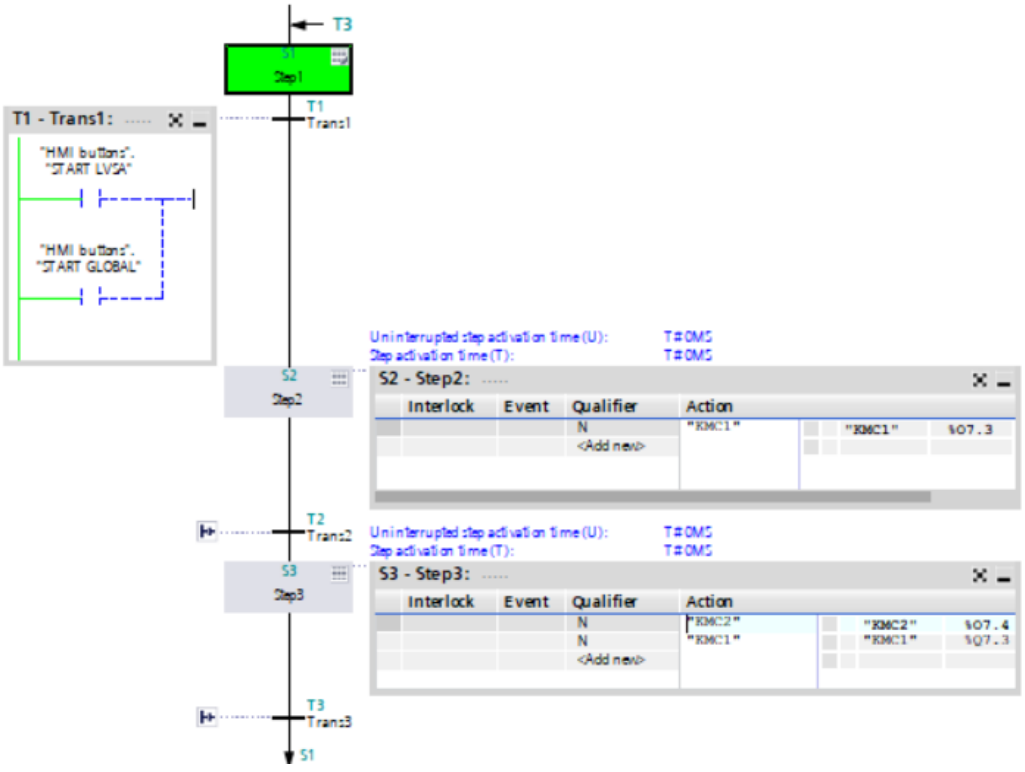


Grafcet MPM

Appendix 6: LVSA Block FB6

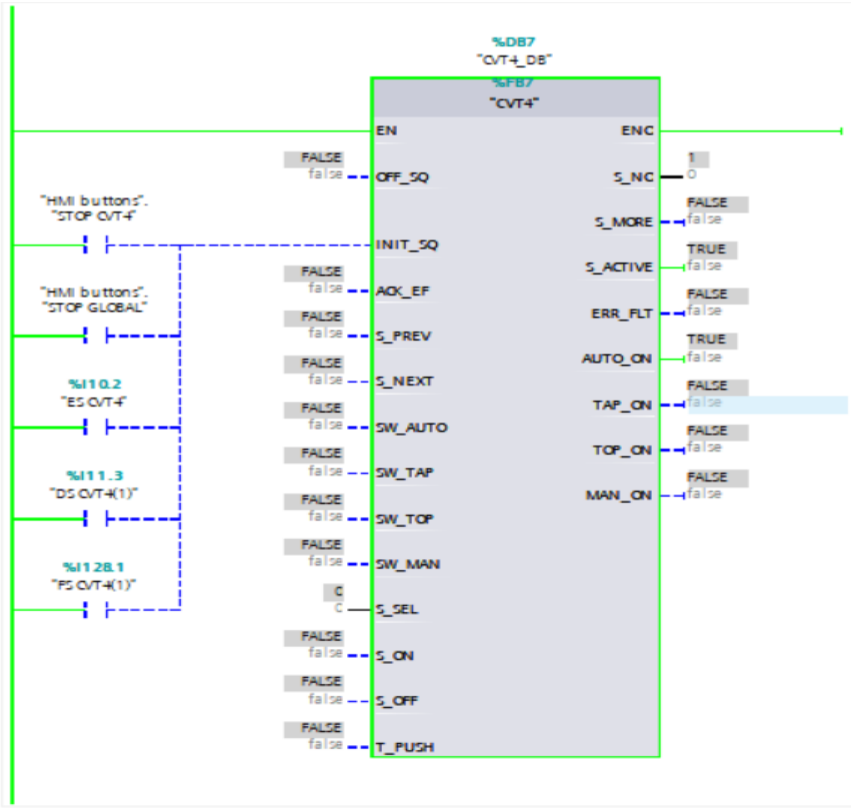


FB6 Call (LVSA) in Main OB1

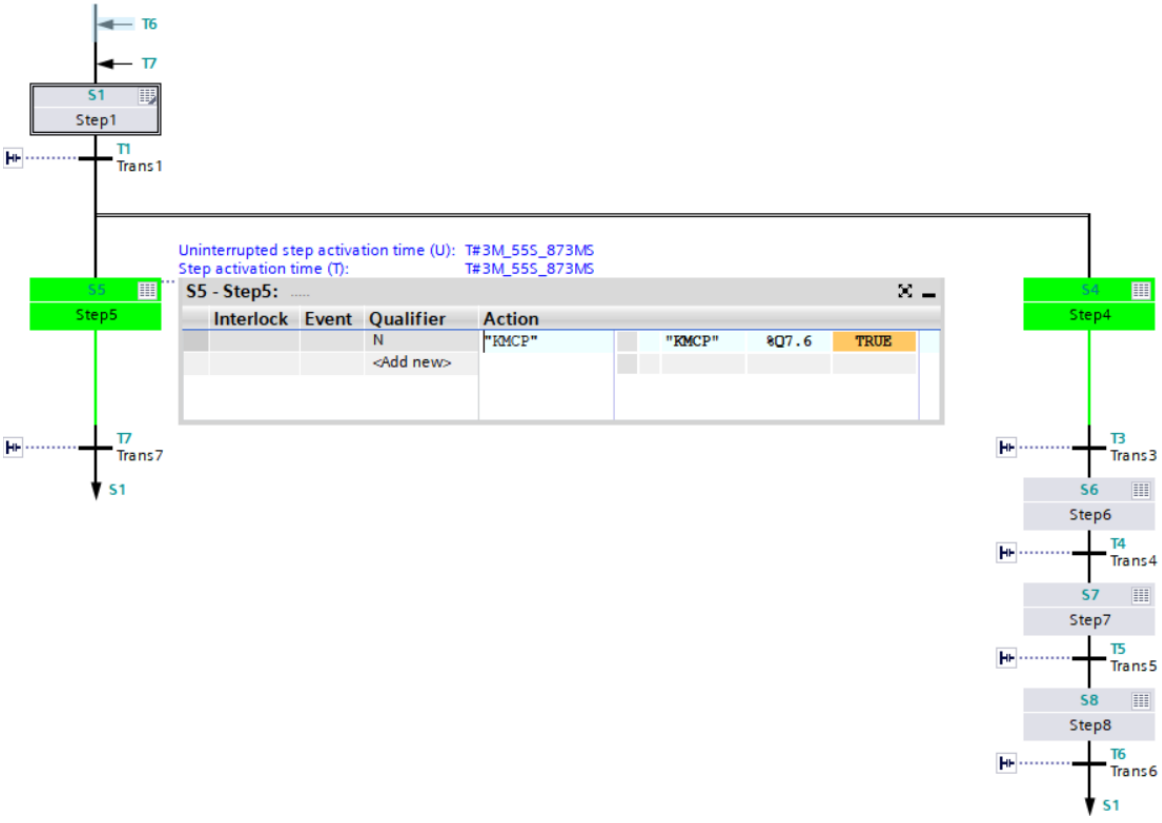


Grafcet LVSA

Appendix 7 : CVT4 Block FB7

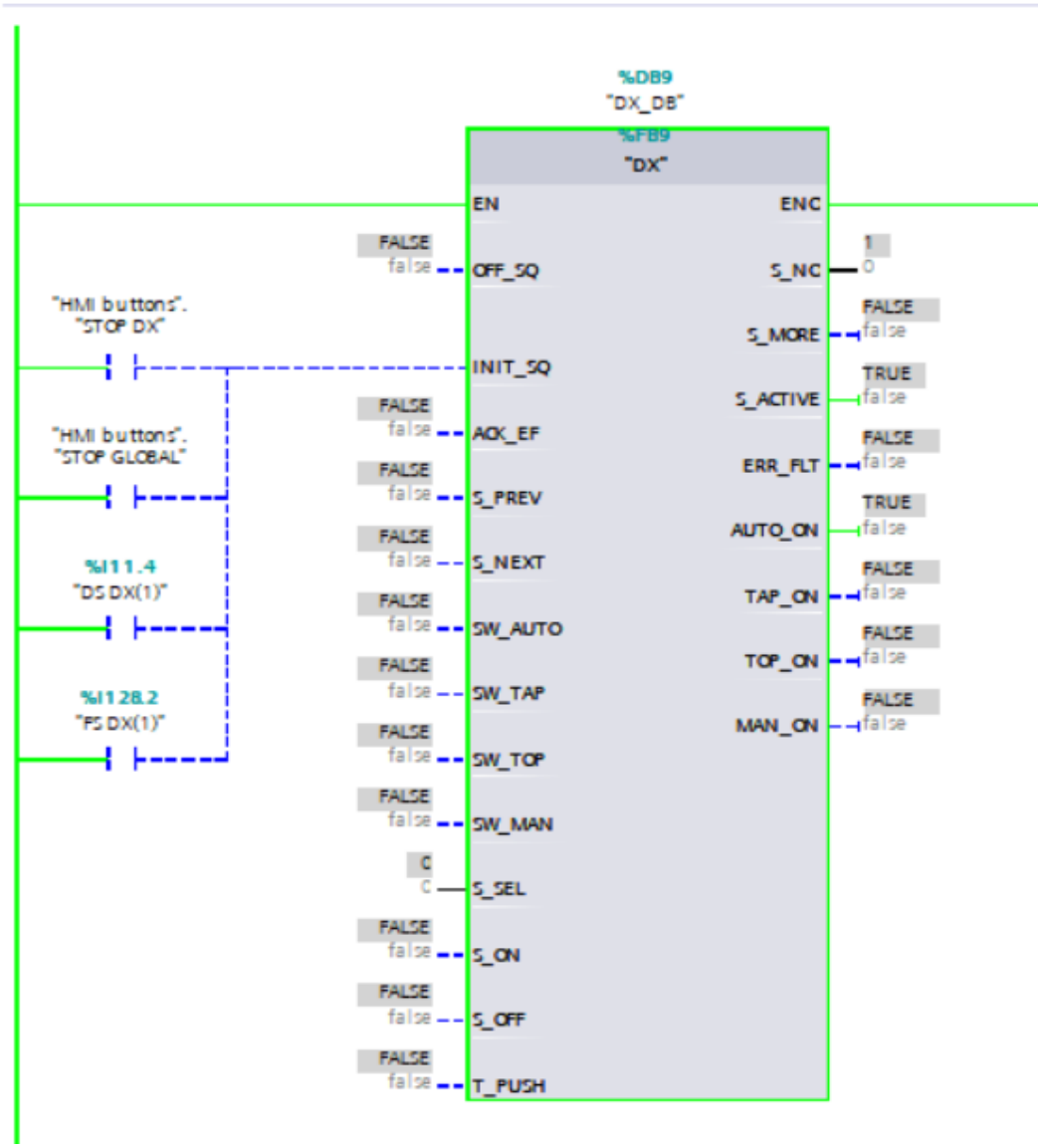


FB7 Call (CVT4) in Main OB1

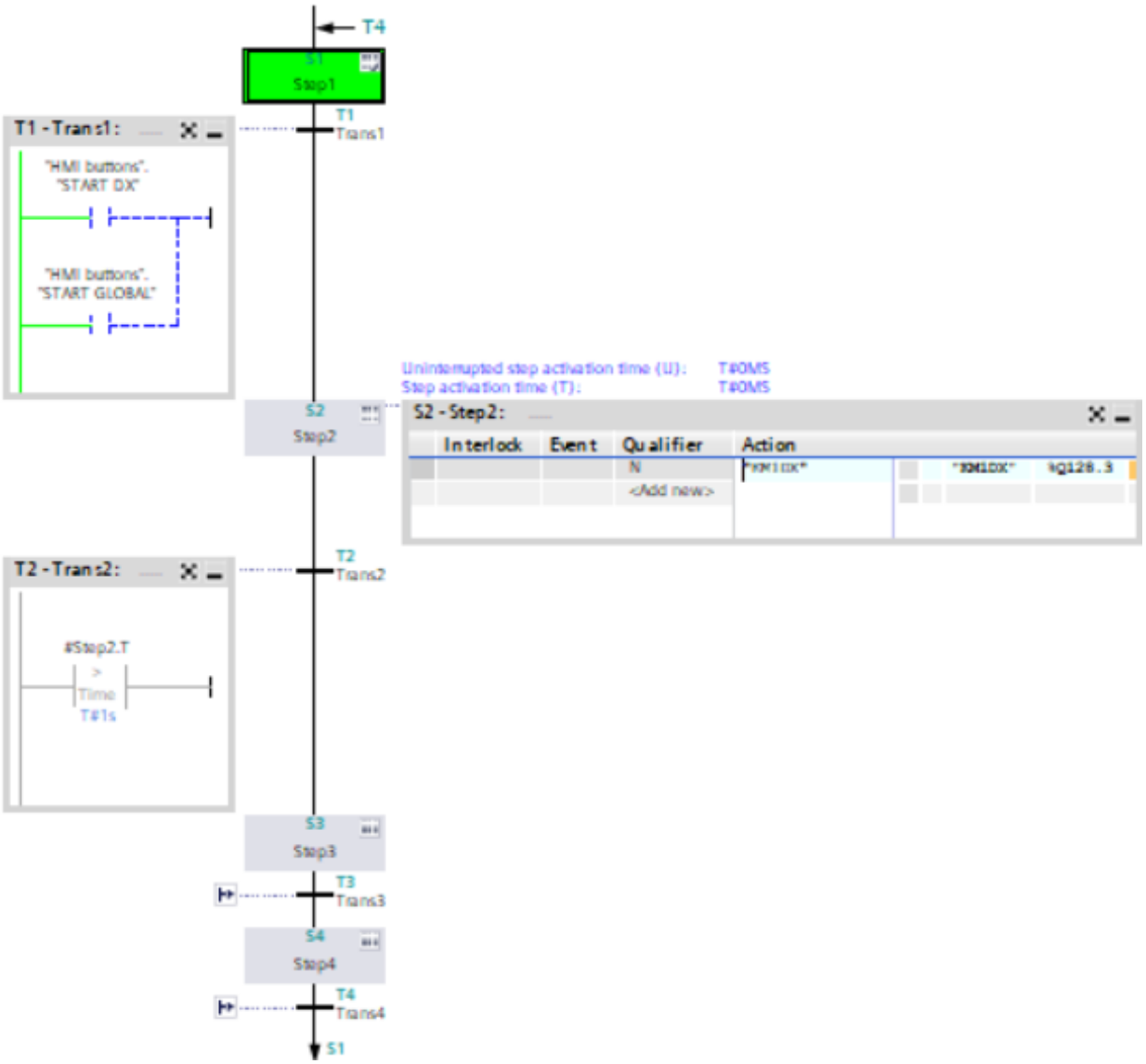


Grafcet CVT4

Appendix 8: DX Block FB9

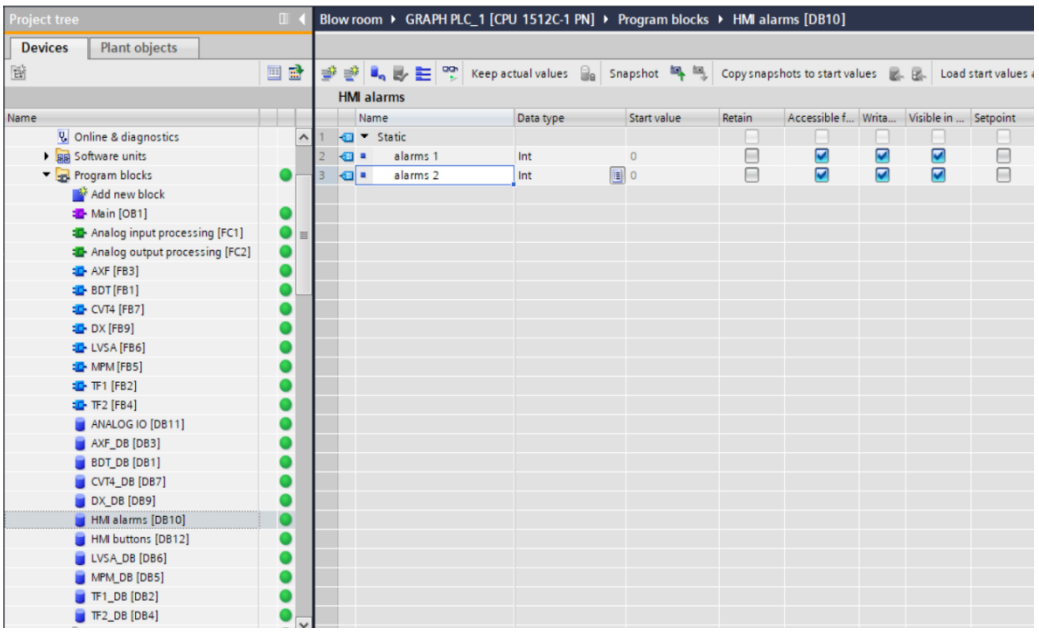


FB9 Call (DX) in Main OB1

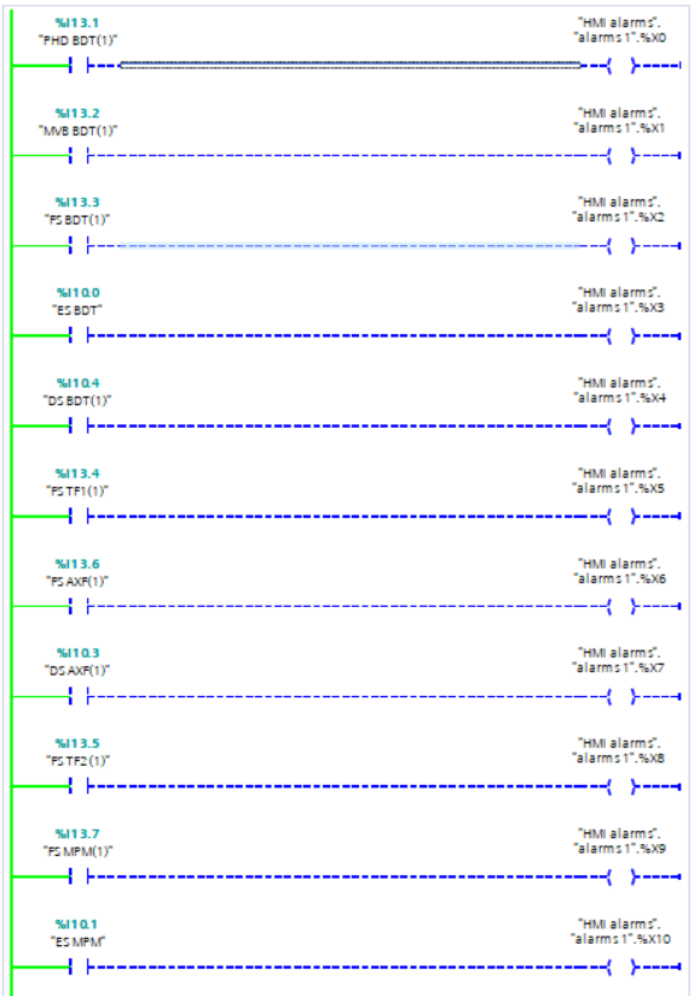

T4
S1
Step1
T1
Trans1
T1 - Trans1:
"HMI buttons".
"START DX"
"HMI buttons".
"START GLOBAL"
Uninterrupted step activation time (U): T#0MS
Step activation time (T): T#0MS
S2
Step2
S2 - Step2:
Interlock
Event
Qualifier
Action
N
<Add new>
T2
Trans2
T2 - Trans2:
#Step2.T
>
Time
T#1s
S3
Step3
T3
Trans3
S4
Step4
T4
Trans4
S1

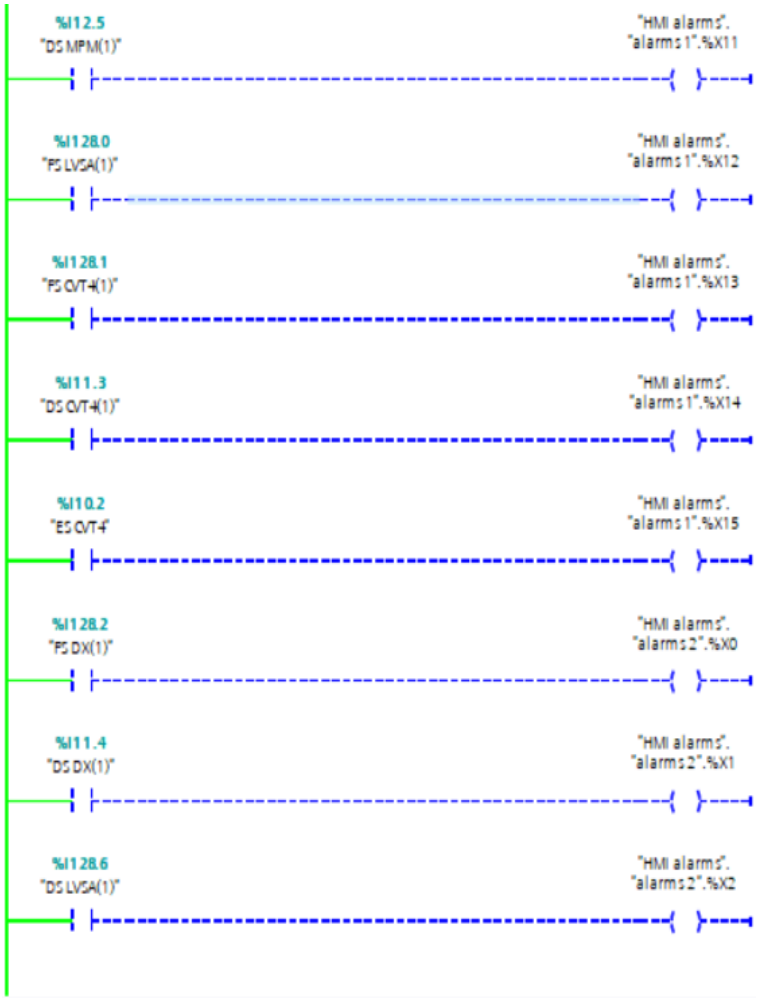
Grafcet DX

Appendix 9: Block DB10



DB10 Call (HMI alarms) in Main OB1



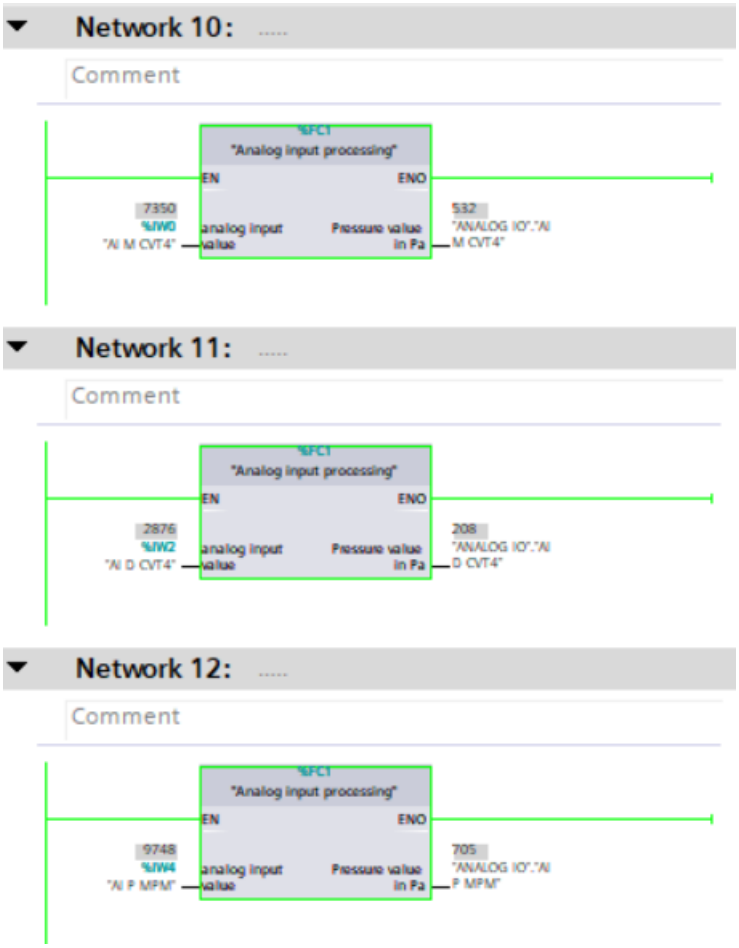


Physical Sensors assigned to DB10 bits

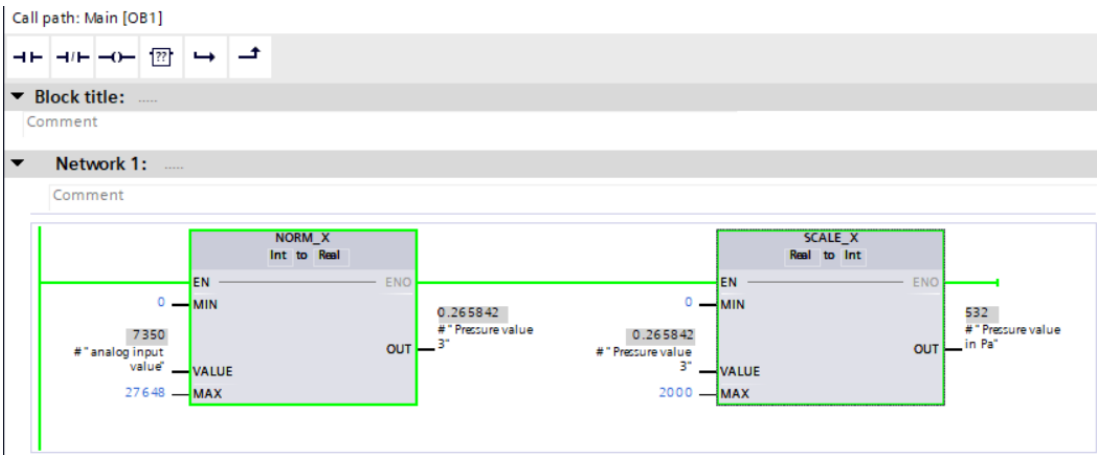
ID	Alarm text	Alarm ...	Trigger tag	Trigge..	Trigger address
1	Human presence detected!	Errors	HMI alarms_alarms 1	0	"HMI alarms"."alarms 1".x0
2	Movement is blocked!	Errors	HMI alarms_alarms 1	1	"HMI alarms"."alarms 1".x1
3	Fire detected!	Errors	HMI alarms_alarms 1	2	"HMI alarms"."alarms 1".x2
4	Emergency stop button is pressed!	Errors	HMI alarms_alarms 1	3	"HMI alarms"."alarms 1".x3
5	Door safety is opened!	Errors	HMI alarms_alarms 1	4	"HMI alarms"."alarms 1".x4
6	Fire detected!	Errors	HMI alarms_alarms 1	5	"HMI alarms"."alarms 1".x5
7	Fire detected!	Errors	HMI alarms_alarms 1	6	"HMI alarms"."alarms 1".x6
8	Door safety is opened!	Errors	HMI alarms_alarms 1	7	"HMI alarms"."alarms 1".x7
9	Fire detected!	Errors	HMI alarms_alarms 1	8	"HMI alarms"."alarms 1".x8
10	Fire detected!	Errors	HMI alarms_alarms 1	9	"HMI alarms"."alarms 1".x9
11	Emergency stop button is pressed!	Errors	HMI alarms_alarms 1	10	"HMI alarms"."alarms 1".x10
12	Door safety is opened!	Errors	HMI alarms_alarms 1	11	"HMI alarms"."alarms 1".x11
13	Fire detected!	Errors	HMI alarms_alarms 1	12	"HMI alarms"."alarms 1".x12
14	Fire detected!	Errors	HMI alarms_alarms 1	13	"HMI alarms"."alarms 1".x13
15	Door safety is opened!	Errors	HMI alarms_alarms 1	14	"HMI alarms"."alarms 1".x14
16	Emergency stop button is pressed!	Errors	HMI alarms_alarms 1	15	"HMI alarms"."alarms 1".x15
17	Fire detected!	Errors	HMI alarms_alarms 2	0	"HMI alarms"."alarms 2".x0
18	Door safety is opened!	Errors	HMI alarms_alarms 2	1	"HMI alarms"."alarms 2".x1
19	Door safety is opened!	Errors	HMI alarms_alarms 2	2	"HMI alarms"."alarms 2".x2

DB10 Call in the HMI Alarms

Appendix 10: Block FC1

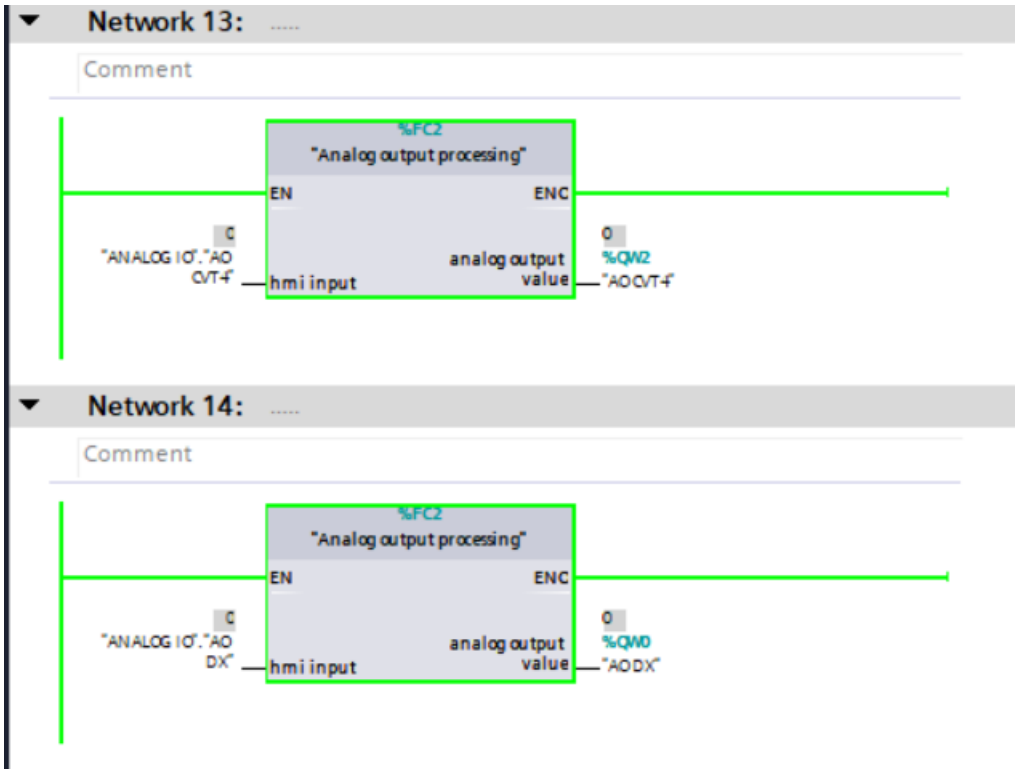


FC1 Call (Analog to Digital Conversion) in OB1

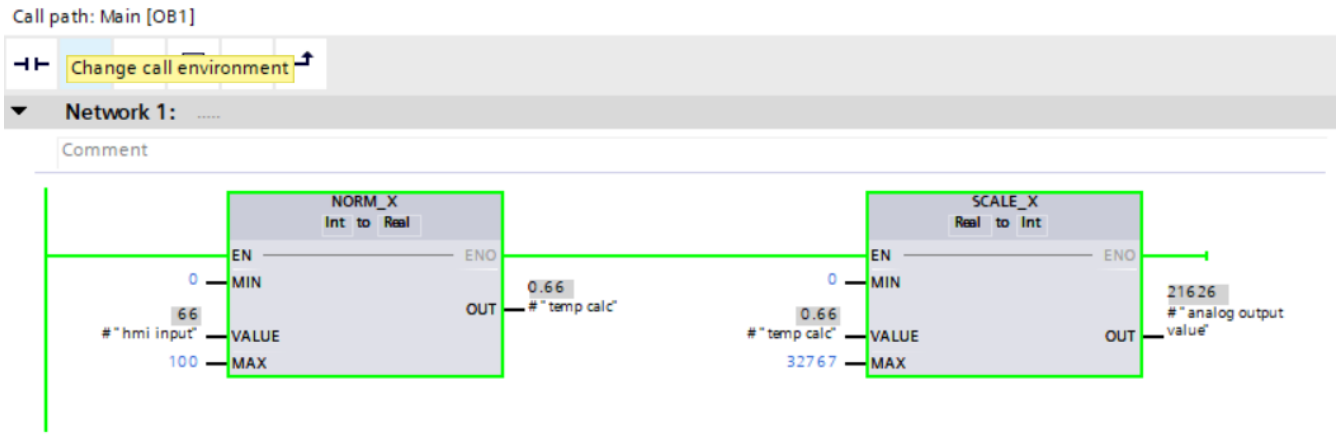


Pressure Conversion: Analog →Digital

Appendix 11: Block FC2



FC2 Call (Digital to Analog Conversion) in OB1



Speed Control Conversion: Digital →Analog

Summary

My end-of-study internship at SITEX Sousse unit, the objective of which is the automation and supervision of a Blow room line. A preliminary study on the function of the machines was carried out in the first place, and the development of the various machine control Grafcets was completed in the second place.

My job was to transform machines based on wired logic into machines that runs on an S7-1500 PLC and program it on the TIA PORTAL programming software.

Finally, a human-machine interface using the HMI in WINCC was created to be able to control the operation of the machines in real time and to intervene in the event of a failure.

Keywords: Grafcets, PLC, S7-1500, HMI, WINCC, TIA PORTAL programming software.